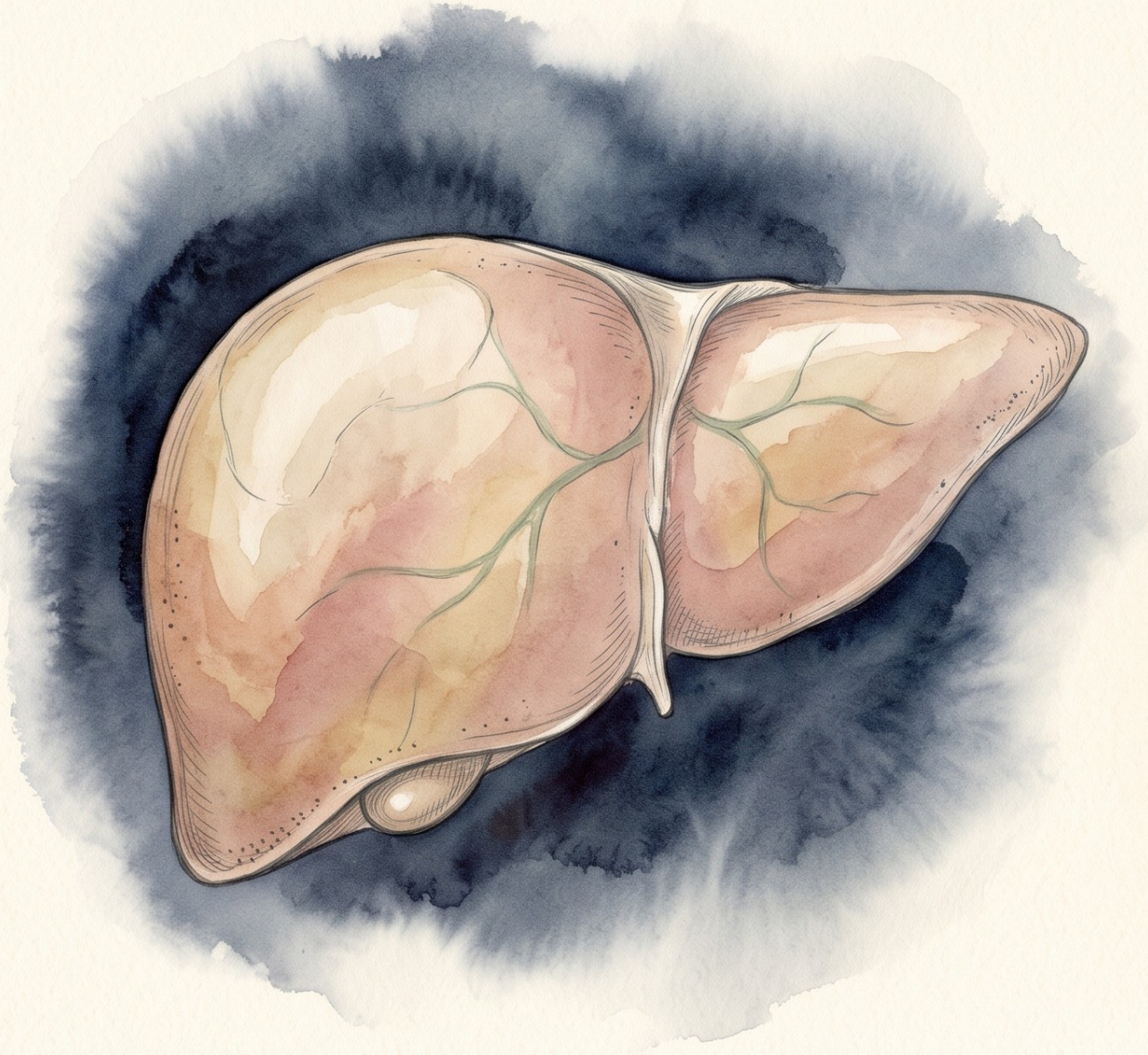


# Acute Liver Failure and ACLF in the ICU

**Mohammad Almeqdadi, MD**
Transplant Hepatologist and Gastroenterologist
Assistant Professor of Medicine and Surgery
X: @mfalmeqdadi

Tufts University | School of Medicine

Slides (PDF)

*Illustrations generated with Google Gemini*

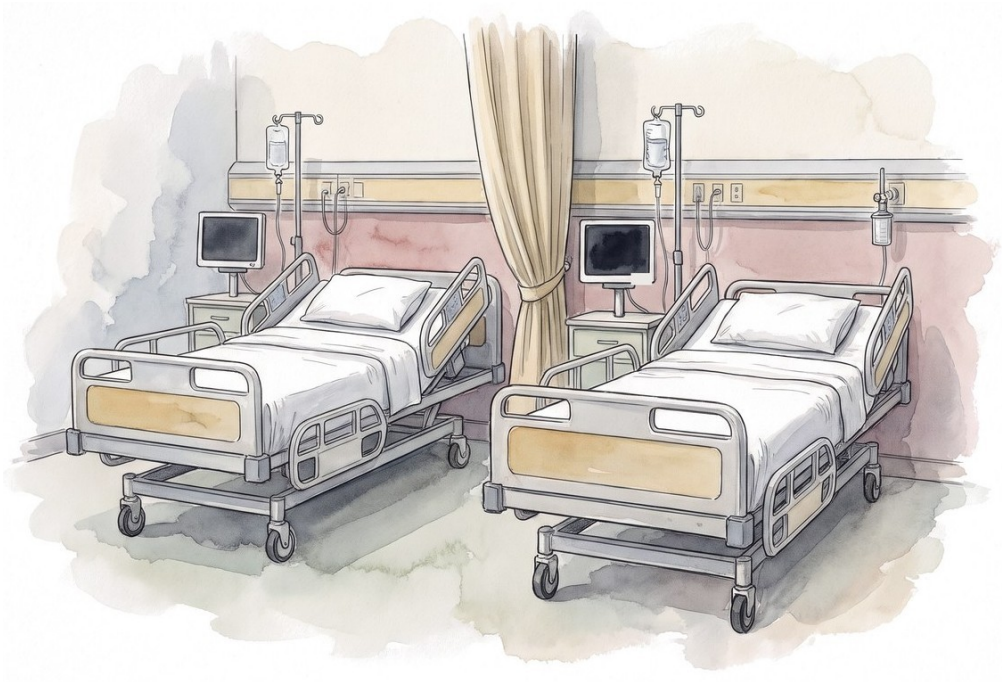
# Objectives

*By the end of this talk you should be able to:*

- Tell acute liver failure apart from severe acute liver injury and from ACLF
- Plan the first hour of care in ALF
- Manage the brain and the ammonia in ALF
- Recognize and grade ACLF, and find the trigger
- Work through AKI in cirrhosis
- Know when to call transplant hepatology

# Two patients, same ICU

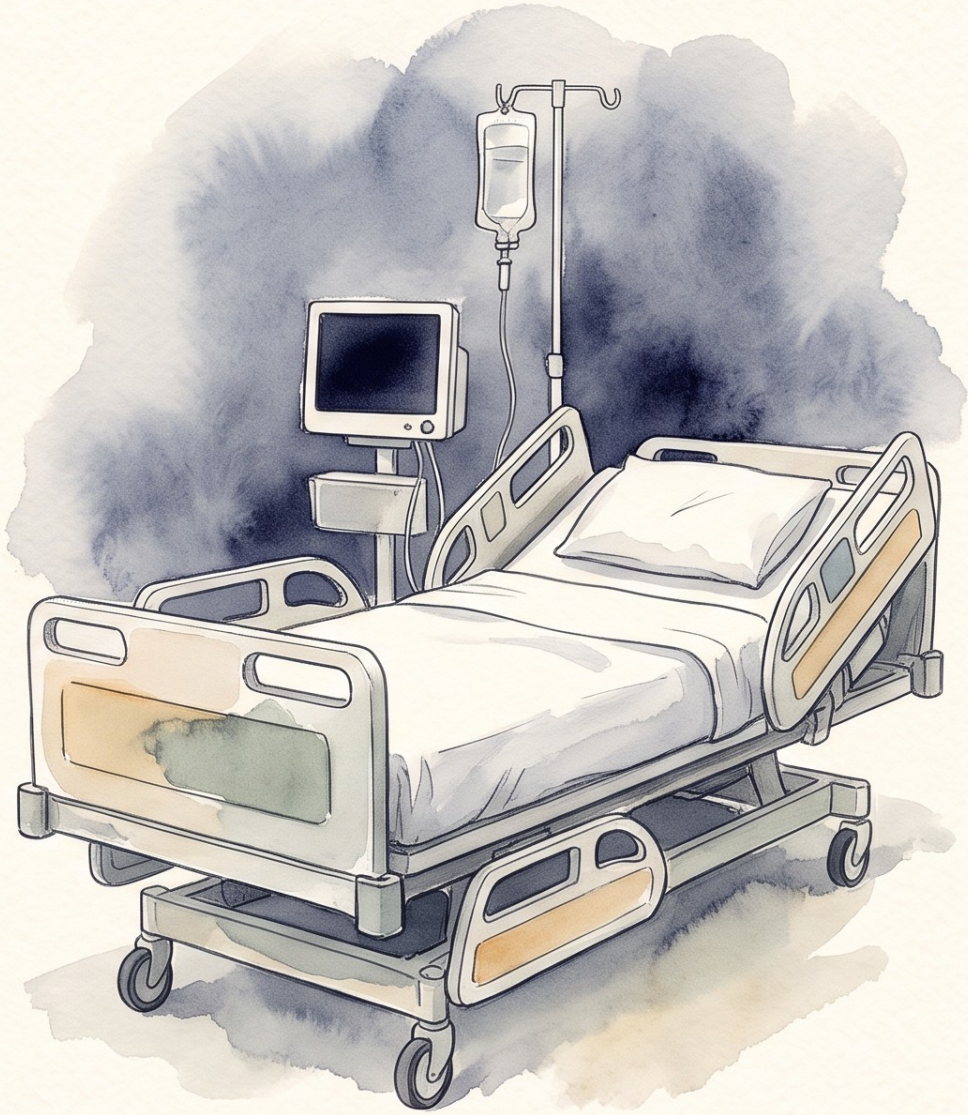
| Bed 4 | Bed 7 |
|---|---|
| 28-year-old woman | 56-year-old man |
| No known liver disease | Alcohol-related cirrhosis with ascites |
| Nausea and vomiting for 4 days | Fever, confusion, low blood pressure |



*Both have a sick liver. Do they have the same disease?*

# Bed 4

28F, 4 days of nausea



# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| ALP | 128 U/L |
| Bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

**Is this acute liver failure?**

# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| ALP | 128 U/L |
| Bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

**Is this acute liver failure?**

**We don't know yet. What's missing?**

# 28F, 4 days of nausea

- Nausea, vomiting and poor intake for 4 days
- No known liver disease, no alcohol use
- "Took something for a toothache"

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| ALP | 128 U/L |
| Bilirubin | 2.9 mg/dL |
| Creatinine | 1.0 mg/dL |

## What's missing?

- **INR** (synthetic function, not enzymes)
- **Mental status** (any encephalopathy?)
- **Time course** (how long since the first symptom?)
- **Is there chronic liver disease underneath?**

# 28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Bilirubin | 2.9 mg/dL |
| **INR** | **3.2** |
| **Glucose** | **54 mg/dL** |

# 28F, 4 days of nausea

- Disoriented to time, asterixis. Still there after the glucose is fixed
- Symptoms started 4 days ago
- Ultrasound: smooth liver, no ascites, patent vessels, normal spleen
- No history or imaging suggesting cirrhosis

| Test | Result |
|---|---|
| AST | 6,820 U/L |
| ALT | 5,940 U/L |
| Bilirubin | 2.9 mg/dL |
| **INR** | **3.2** |
| **Glucose** | **54 mg/dL** |

**This is acute liver failure**

# Acute liver failure

- **INR ≥ 1.5** and **any degree of encephalopathy**
- Illness of less than 26 weeks
- No pre-existing cirrhosis
  - AASLD and ACG: chronic liver disease without cirrhosis still allows ALF. EASL: no prior liver disease at all
  - Wilson disease, autoimmune hepatitis and vertically acquired HBV can still count if recognized within 26 weeks

| Jaundice to encephalopathy | Hyperacute (< 1 week) | Acute (1 to 4 weeks) | Subacute (4 to 12 weeks) |
|---|---|---|---|
| Typical causes | Acetaminophen, HAV, HEV, ischemia | HBV | DILI, AIH, unknown cause |
| Cerebral edema | **Common** | Less common | Uncommon |
| Spontaneous recovery | More likely | Intermediate | **Less likely** |

*EASL ALF guideline 2017 [PMID: 28417882]; ACG ALF 2023 [PMID: 37377263]; Management of ALF: update 2022 [PMID: 36001996]*

# Same patient, but wide awake

*Same labs and INR 2.4, but fully oriented with no asterixis*

**What do you call this, and what do you do now?**



# Same patient, but wide awake

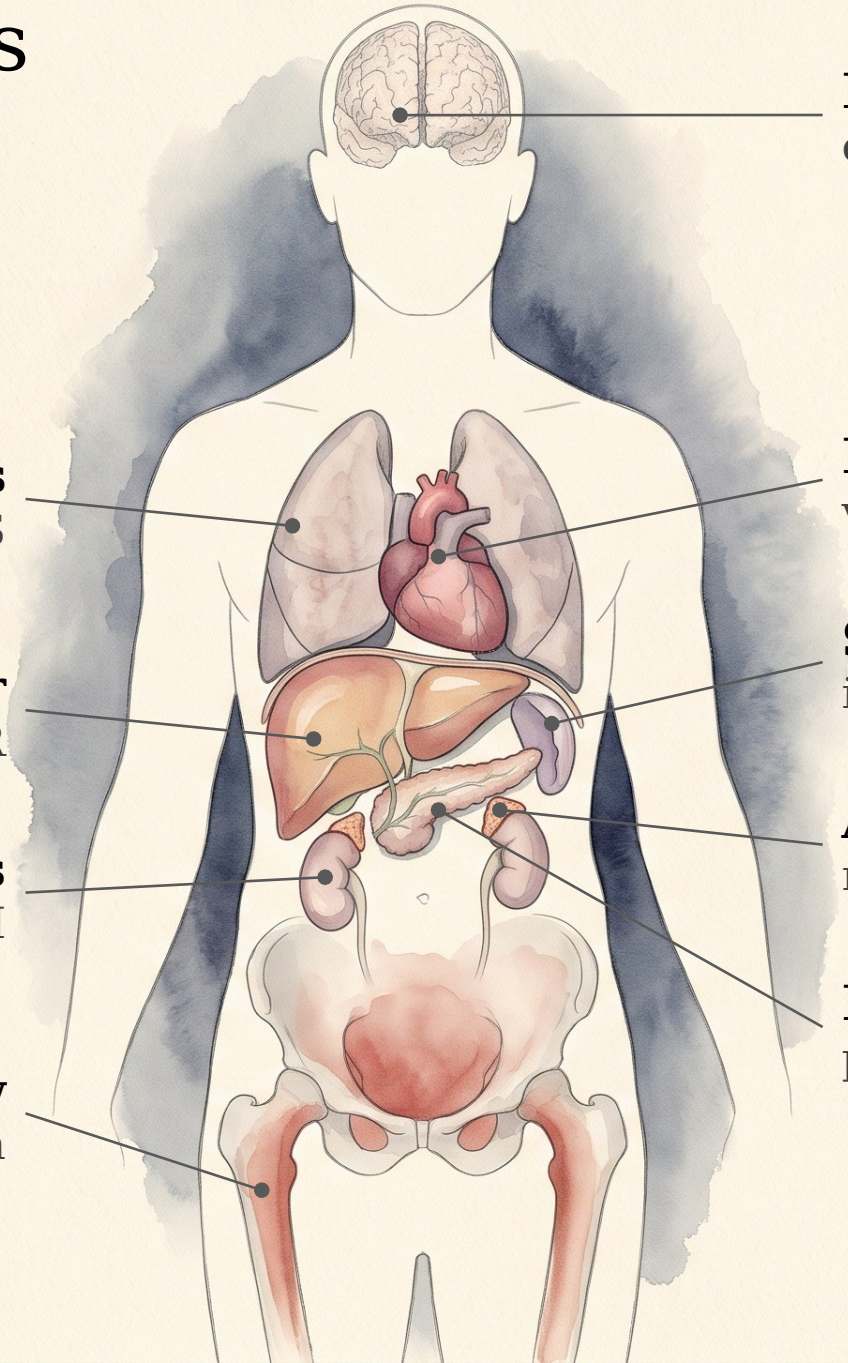
*Same labs and INR 2.4, but fully oriented with no asterixis*

- **Severe acute liver injury**, not ALF yet
- No encephalopathy, so it doesn't meet the definition
- But it can become ALF within hours
  - Check glucose often and examine the patient every few hours
- And the opposite trap:
  - Hypoglycemia, sedatives, sepsis, alcohol withdrawal and seizures can all look like encephalopathy

# Which other organs fail in ALF?



# Which other organs fail in ALF?



**Brain**
encephalopathy, cerebral edema

**Lungs**
ARDS

**Heart and vessels**
vasodilatory shock

**Liver**
hypoglycemia, lactate, INR

**Spleen and immune system**
infection

**Kidneys**
AKI

**Adrenals**
relative insufficiency

**Pancreas**
pancreatitis

**Bone marrow**
suppression

*Figure content after Bernal W, Wendon J. N Engl J Med 2013;369:2525-34*

28F, 4 days of nausea

**What's your next step?**

# Two things at the same time

| Keep her alive | Find the cause |
|---|---|
| ICU bed | Acetaminophen level |
| Glucose every 1 to 2 hours | Viral, autoimmune, Wilson, toxicology |
| Volume, perfusion, lactate | Doppler ultrasound |
| Neuro checks | Pregnancy test |

**Both start in the first hour**

# Which cause changes treatment right away?

| If you suspect | Clues | Send | Don't wait to |
|---|---|---|---|
| Acetaminophen | AST/ALT in the thousands, OTC use | Level, combination products | |
| Herpes simplex | Fever, very high AST/ALT, little jaundice | HSV PCR (blood) | |
| Hepatitis B | Endemic area, chemo or immunosuppression | HBsAg, anti-HBc IgM, DNA | |
| Autoimmune hepatitis | Woman, other autoimmune disease | ANA, ASMA, IgG | |
| Wilson disease | Under 40, Coombs-negative hemolysis | Ceruloplasmin, urine copper | |
| AFLP, HELLP | Third trimester, high BP, low platelets | hCG | |
| Budd-Chiari | Pain, ascites, big liver | Doppler or CT | |
| Ischemic hepatitis | Shock or arrest, very high LDH | Echo | |
| Amanita | Wild mushrooms, diarrhea 6-24 h later | Poison control | |

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]*

# Which cause changes treatment right away?

| If you suspect | Clues | Send | Don't wait to |
|---|---|---|---|
| Acetaminophen | AST/ALT in the thousands, OTC use | Level, combination products | Start NAC |
| Herpes simplex | Fever, very high AST/ALT, little jaundice | HSV PCR (blood) | Start IV acyclovir |
| Hepatitis B | Endemic area, chemo or immunosuppression | HBsAg, anti-HBc IgM, DNA | Entecavir or tenofovir |
| Autoimmune hepatitis | Woman, other autoimmune disease | ANA, ASMA, IgG | Steroids with hepatology |
| Wilson disease | Under 40, Coombs-negative hemolysis | Ceruloplasmin, urine copper | Urgent transplant |
| AFLP, HELLP | Third trimester, high BP, low platelets | hCG | Obstetrics: delivery |
| Budd-Chiari | Pain, ascites, big liver | Doppler or CT | Anticoagulate; TIPS |
| Ischemic hepatitis | Shock or arrest, very high LDH | Echo | Fix the circulation |
| Amanita | Wild mushrooms, diarrhea 6-24 h later | Poison control | NAC, silibinin |

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]; EASL AIH 2025 [PMID: 40348684]*

# 28F, 4 days of nausea

- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown

**Can you use the nomogram?**



# 28F, 4 days of nausea

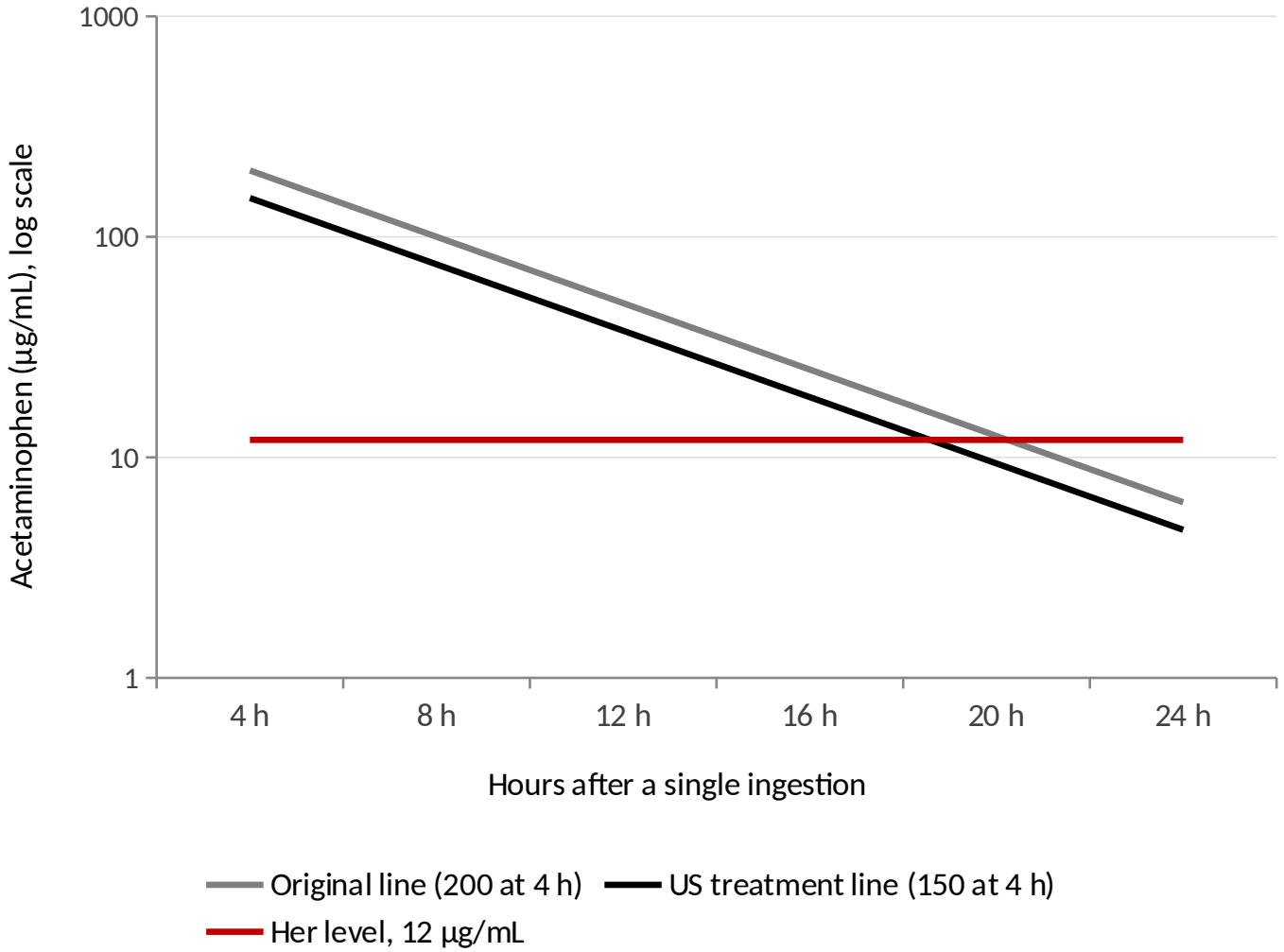
- Roommate: "She was taking a lot of the Tylenol with codeine, plus cold medicine, for 3 days"
- Acetaminophen level: **12 µg/mL**, time of last dose unknown



## Can you use the nomogram?

- **No.** Rumack-Matthew is for a single acute ingestion with a known time
- Repeated supratherapeutic use and late presentation: the level can be low or undetectable
- **Start NAC now**

# The Rumack-Matthew nomogram




- **One acute ingestion at a known time: it tells you when NOT to give NAC**
- Late, repeated or unknown-time ingestion: a low level doesn't rule out liver injury

**If she had taken it all at once**

- 8 hours: 12 is below the line, no NAC
- 20 hours: above the line, NAC
- But it was 3 days of repeated doses. **There is no time zero**. Give NAC

*Rumack BH, Matthew H. Pediatrics 1975; US/Canada acetaminophen consensus 2023 [PMID: 37552484]*

# NAC for every kind of acute liver failure?

Published in final edited form as:
*Gastroenterology*. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

**WM Lee**[1], **LS Hynan**[1], **L Rossaro**[2], **RJ Fontana**[3], **RT Stravitz**[4], **AM Larson**[5], **TL Davern II**[6], **NG Murray**[7], **T McCashland**[8], **JS Reisch**[1], **PR Robuck**[9], and **the Acute Liver Failure Study Group**

**What do you think they found?**

# NAC for every kind of acute liver failure?

Published in final edited form as:
*Gastroenterology*. 2009 September ; 137(3): 856–864.e1. doi:10.1053/j.gastro.2009.06.006.

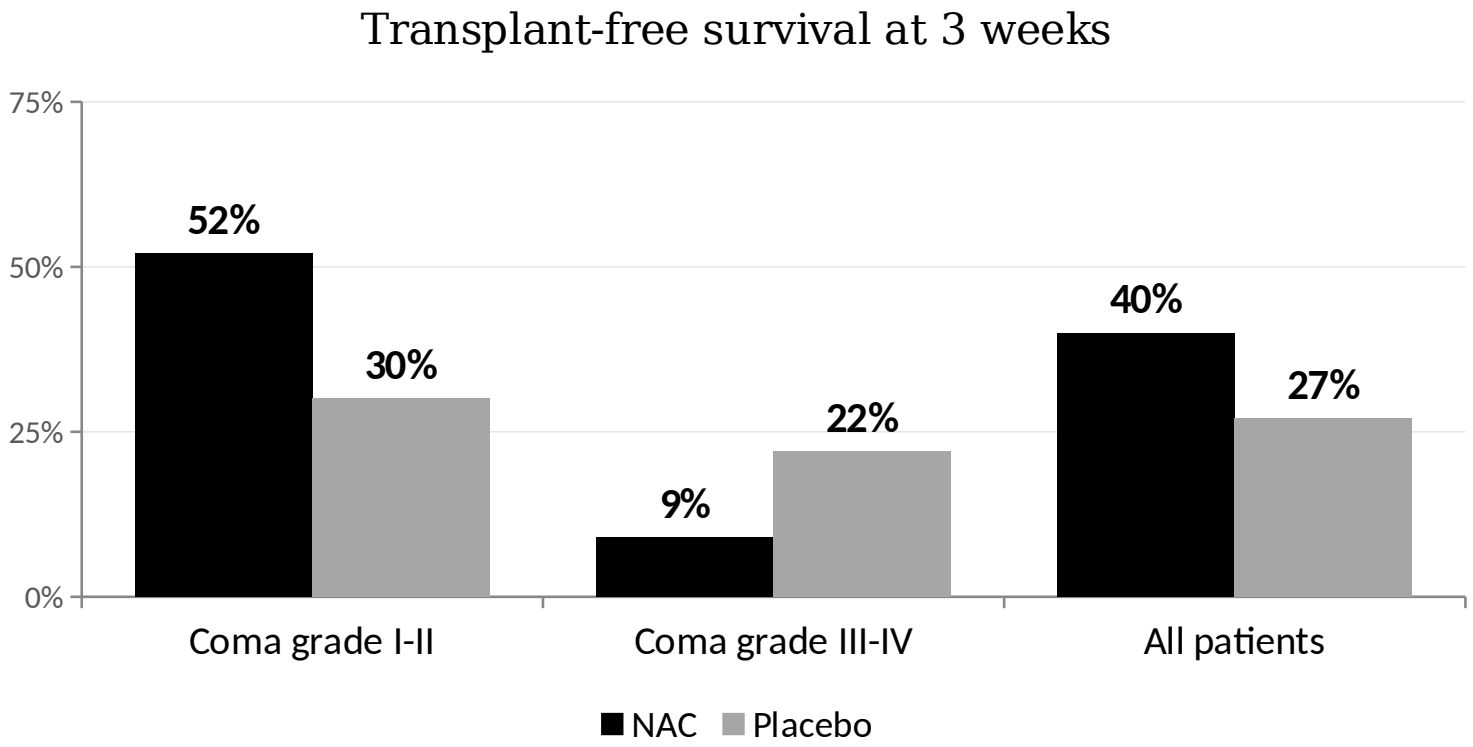
## INTRAVENOUS *N*-ACETYLCYSTEINE IMPROVES TRANSPLANT-FREE SURVIVAL IN EARLY STAGE NON-ACETAMINOPHEN ACUTE LIVER FAILURE

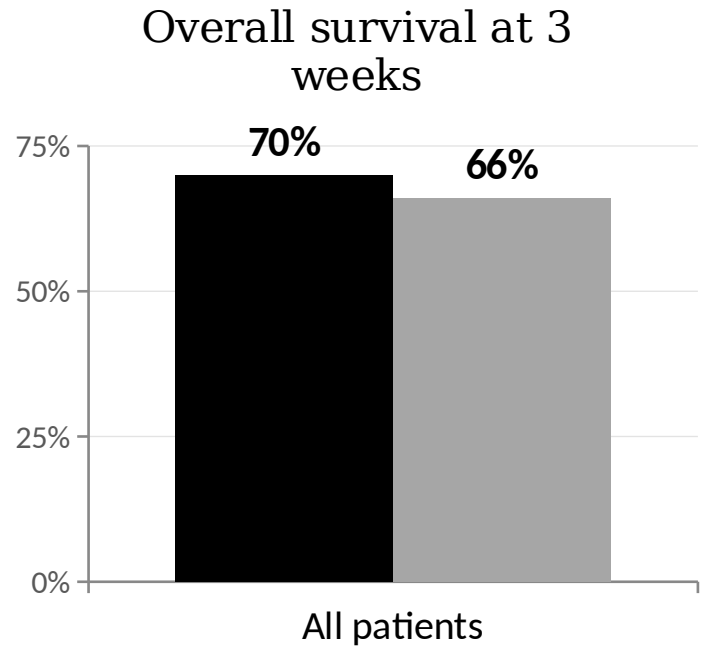
**WM Lee**[1], **LS Hynan**[1], **L Rossaro**[2], **RJ Fontana**[3], **RT Stravitz**[4], **AM Larson**[5], **TL Davern II**[6], **NG Murray**[7], **T McCashland**[8], **JS Reisch**[1], **PR Robuck**[9], and **the Acute Liver Failure Study Group**

**Conclusions—**Intravenous NAC improves transplant-free survival in patients with early stage non-acetaminophen related acute liver failure. Patients with advanced coma grades do not benefit from NAC and typically require emergency liver transplantation. (ClinicalTrials.gov number NCT00004467)

# NAC for every kind of acute liver failure?

*Adult non-acetaminophen ALF, randomized, NAC vs placebo, 173 patients*




**Reasonable in early non-acetaminophen ALF. Don't extrapolate to advanced coma, children or ACLF**

*Lee WM et al. Gastroenterology 2009 [PMID: 19524577]*

# 28F, 4 days of nausea

| | Admission | Hour 12 |
|---|---|---|
| Mental status | Disoriented, asterixis | **Responds only to voice (grade III)** |
| Ammonia (arterial) | Not sent | **182 µmol/L** |
| Urine output | OK | 15 mL/h for 4 hours |
| Lactate (after 3 L) | 2.1 mmol/L | 4.6 mmol/L |
| Creatinine | 1.0 mg/dL | 1.3 mg/dL |
| Sodium | 136 mmol/L | 133 mmol/L |
| Glucose | 54 mg/dL | Needs D10 |

## What worries you most?

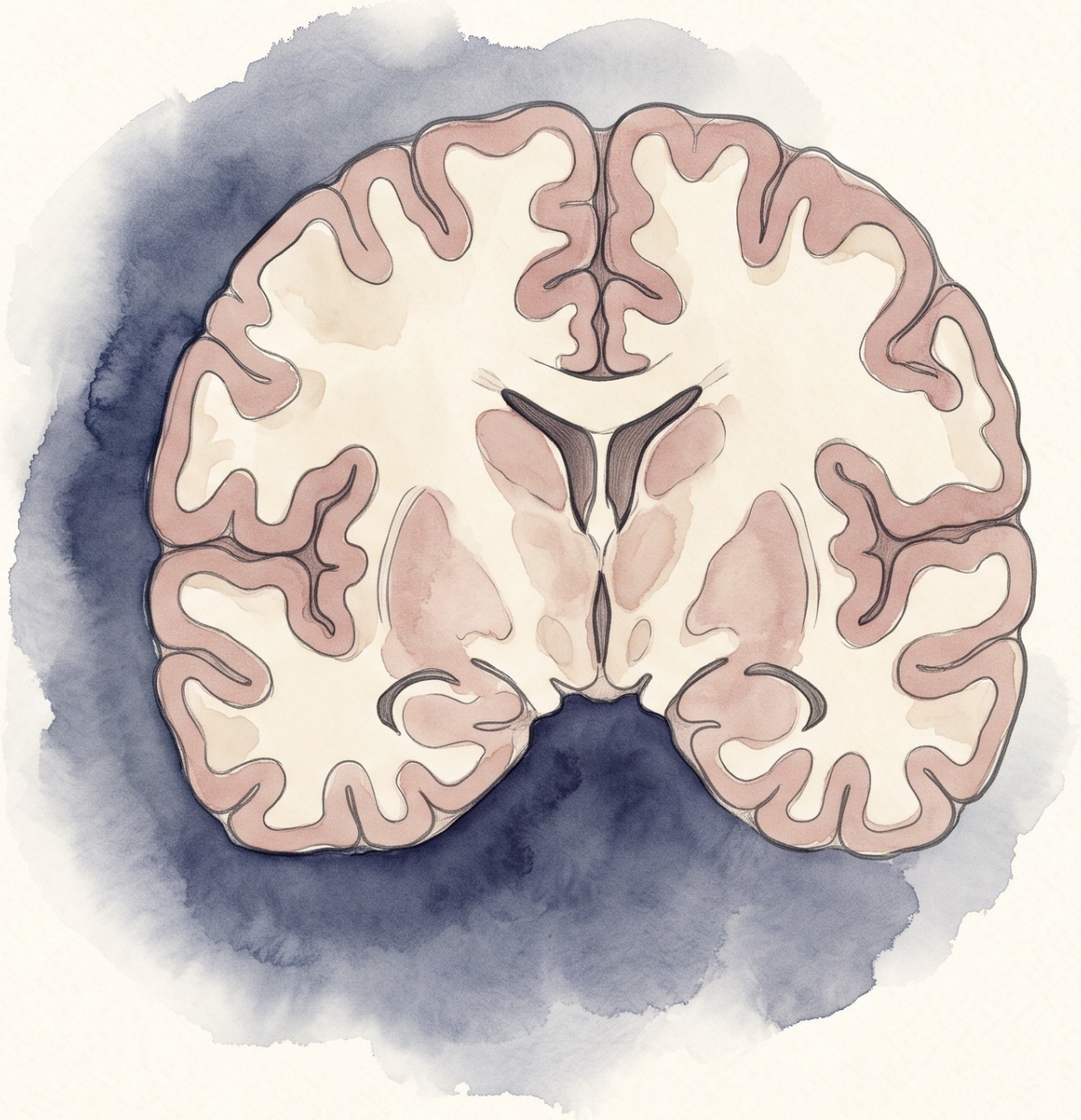
# 28F, 4 days of nausea

| | Admission | Hour 12 |
|---|---|---|
| Mental status | Disoriented, asterixis | **Responds only to voice (grade III)** |
| Ammonia (arterial) | Not sent | **182 µmol/L** |
| Urine output | OK | 15 mL/h for 4 hours |
| Lactate (after 3 L) | 2.1 mmol/L | 4.6 mmol/L |
| Creatinine | 1.0 mg/dL | 1.3 mg/dL |
| Sodium | 136 mmol/L | 133 mmol/L |
| Glucose | 54 mg/dL | Needs D10 |

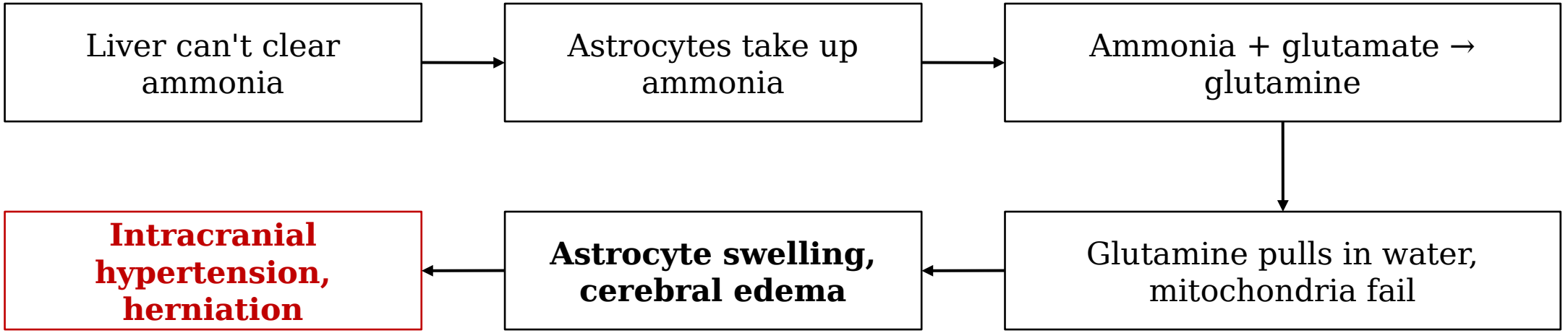
## Her brain

*When do you intubate? Do you wait for the creatinine before dialysis?*

# Ammonia and the brain



# Why does the brain swell?



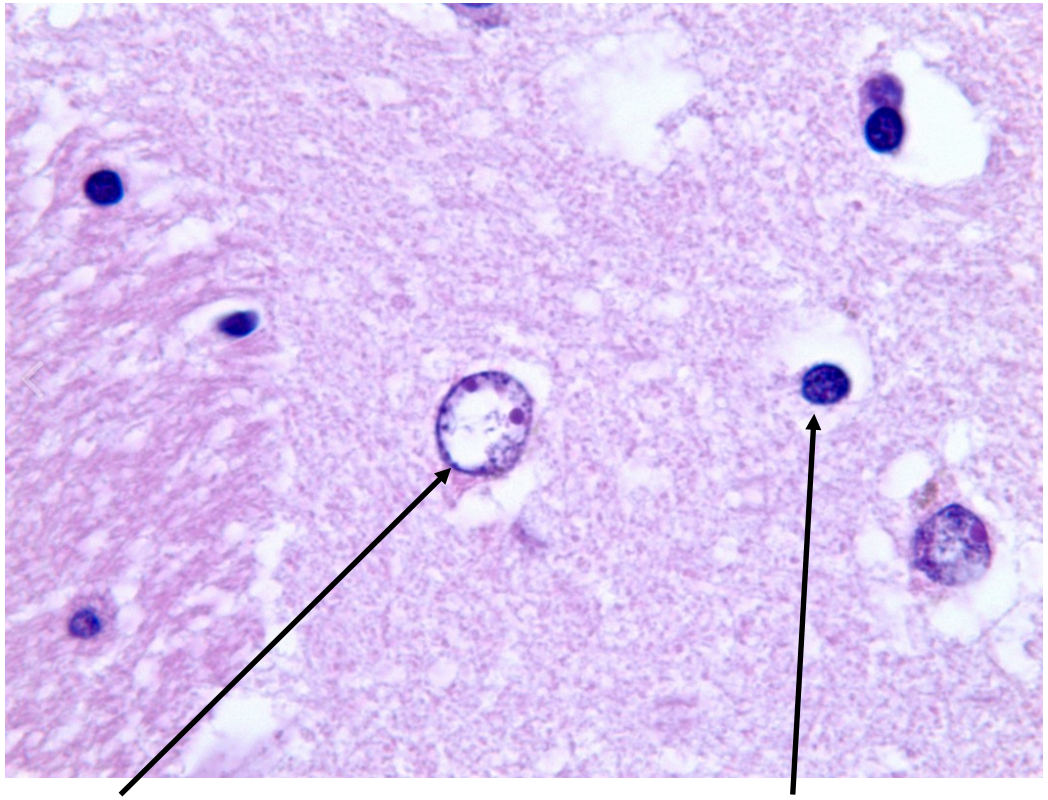

- **Makes it worse:** inflammation and infection, loss of autoregulation, fever, low sodium
- **Highest risk:** hyperacute course, young, grade III-IV encephalopathy, ammonia persistently above 150 to 200 µmol/L, AKI, vasopressors

*Update 2022 [PMID: 36001996]; practical update 2024 [PMID: 39170946]*

# Same ammonia, different brain

**Why do patients with cirrhosis rarely get cerebral edema, even with a high ammonia?**

# Same ammonia, different brain



**Alzheimer type II astrocyte:** big, pale nucleus

**Normal glial nucleus:** small and dark

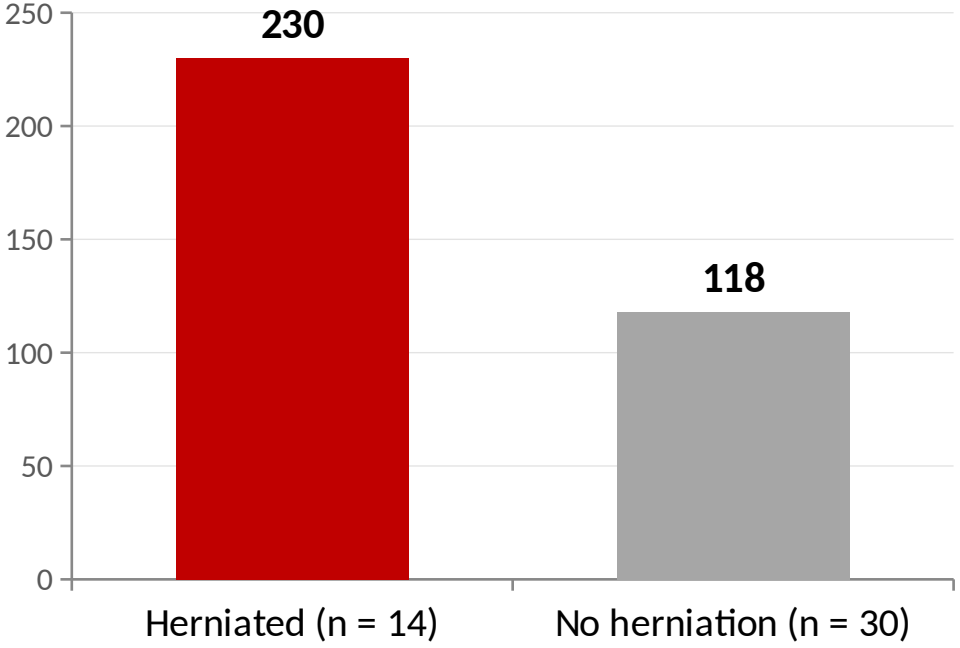
- Astrocytes clear ammonia by turning it and glutamate into glutamine
- Glutamine pulls water into the cell
- **Cirrhosis:** over months the astrocyte pushes out other solutes (myo-inositol, taurine) to make room. It swells only a little. This is the cell you see under the microscope
- **ALF:** the ammonia comes in days. No time to adapt, so the astrocytes swell and the brain swells with them

**Cerebral edema is common in hyperacute ALF and rare in cirrhosis**

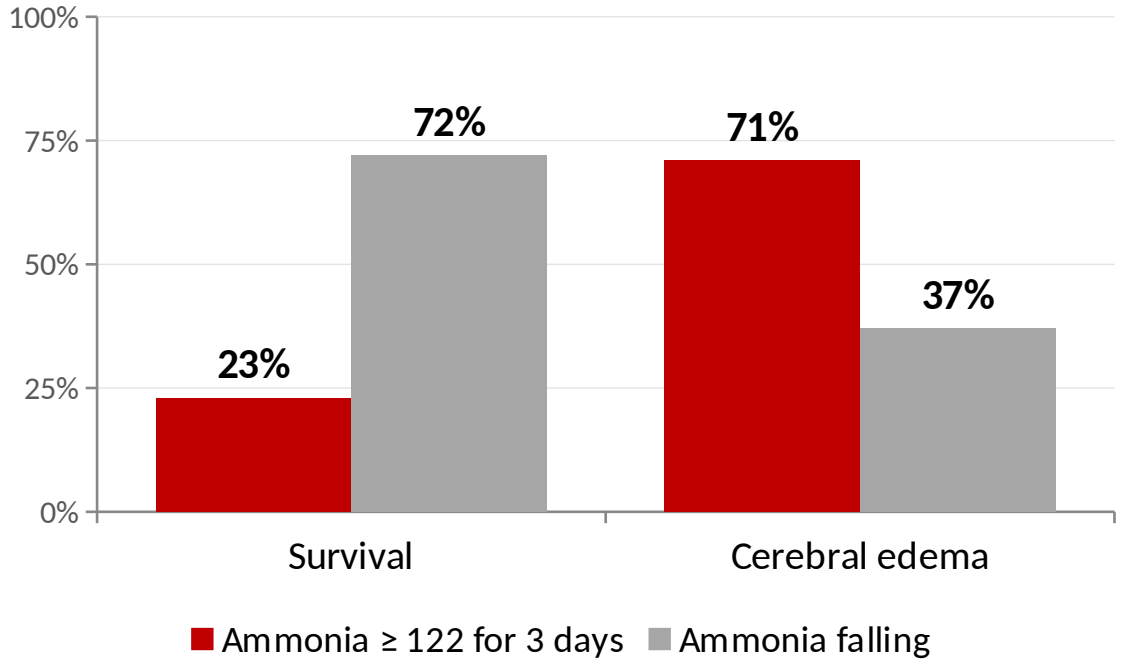



# Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L




Kumar 2012: 295 patients



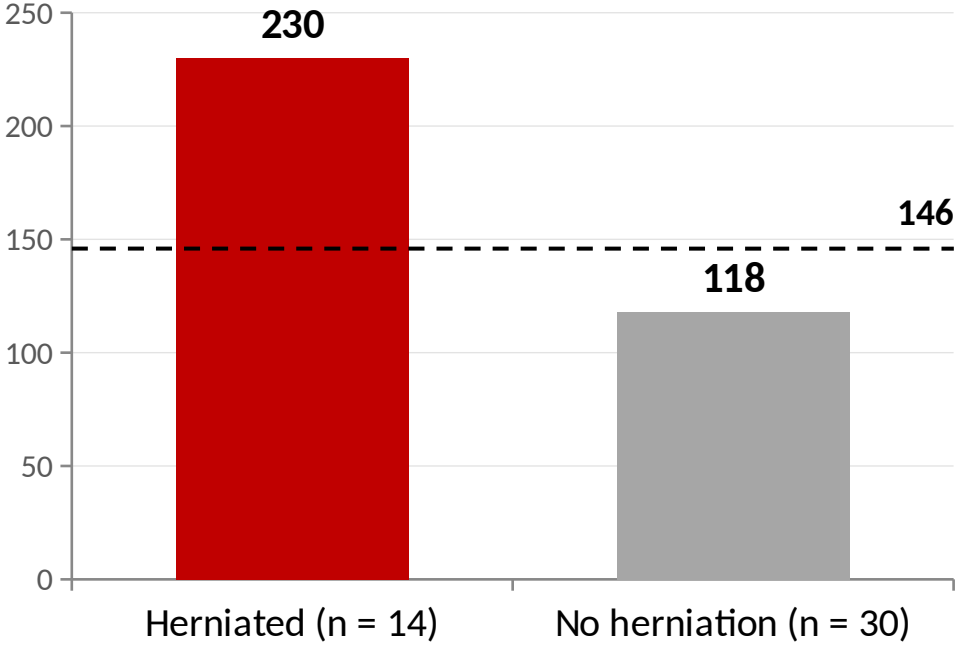

**So what number worries you?**

*Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix*

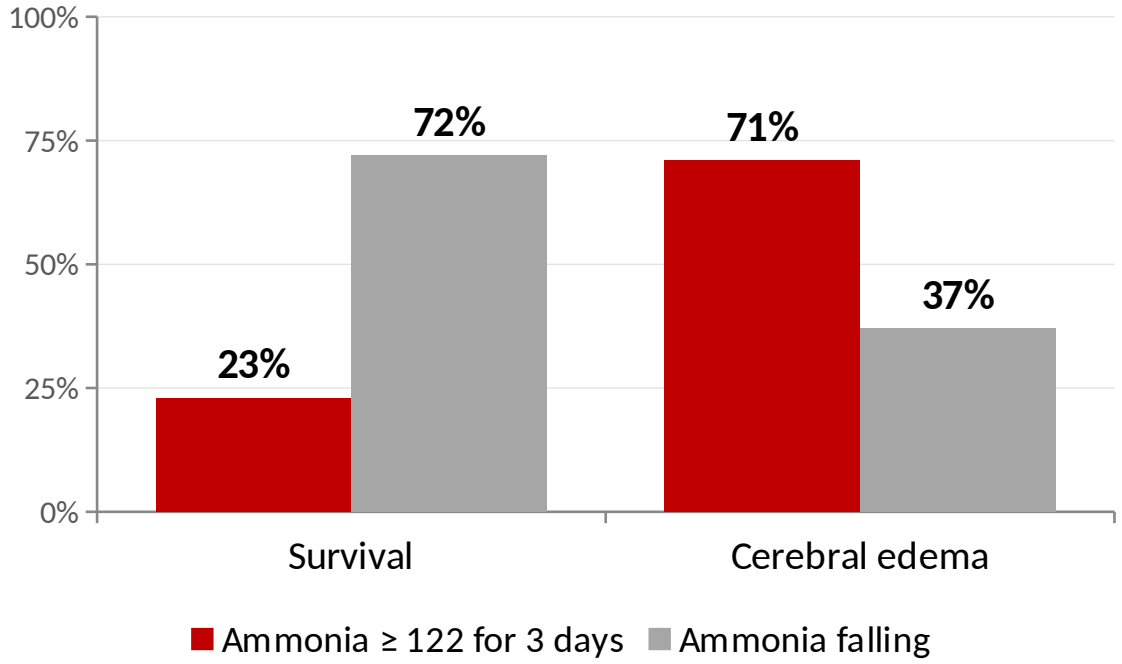



# Ammonia: how high is too high?

Clemmesen 1999: ammonia, µmol/L




Kumar 2012: 295 patients




**Above 150 to 200 and not falling: worry about the brain**

*No single level rules ICH in or out. Read it with the HE grade and the trend*

*Clemmesen JO et al. Hepatology 1999 [PMID: 10051463]; Kumar R et al. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]; more studies in the appendix*

# Lactulose for her?

*Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it*

**Yes or no?**



# Lactulose for her?

*Hour 12: ammonia 182 µmol/L, grade III. In a cirrhotic we'd already be giving it*

| Why people give it | Why not |
|---|---|
| Ammonia drives brain swelling | **No proven benefit in ALF (ALFSG)** |
| It's what we do in cirrhosis | Distended bowel gets in the way at transplant |
| Cheap; AASLD 2011 allowed it early | Diarrhea: free water loss, sodium swings |
| | Liquid by mouth or tube: aspiration risk |

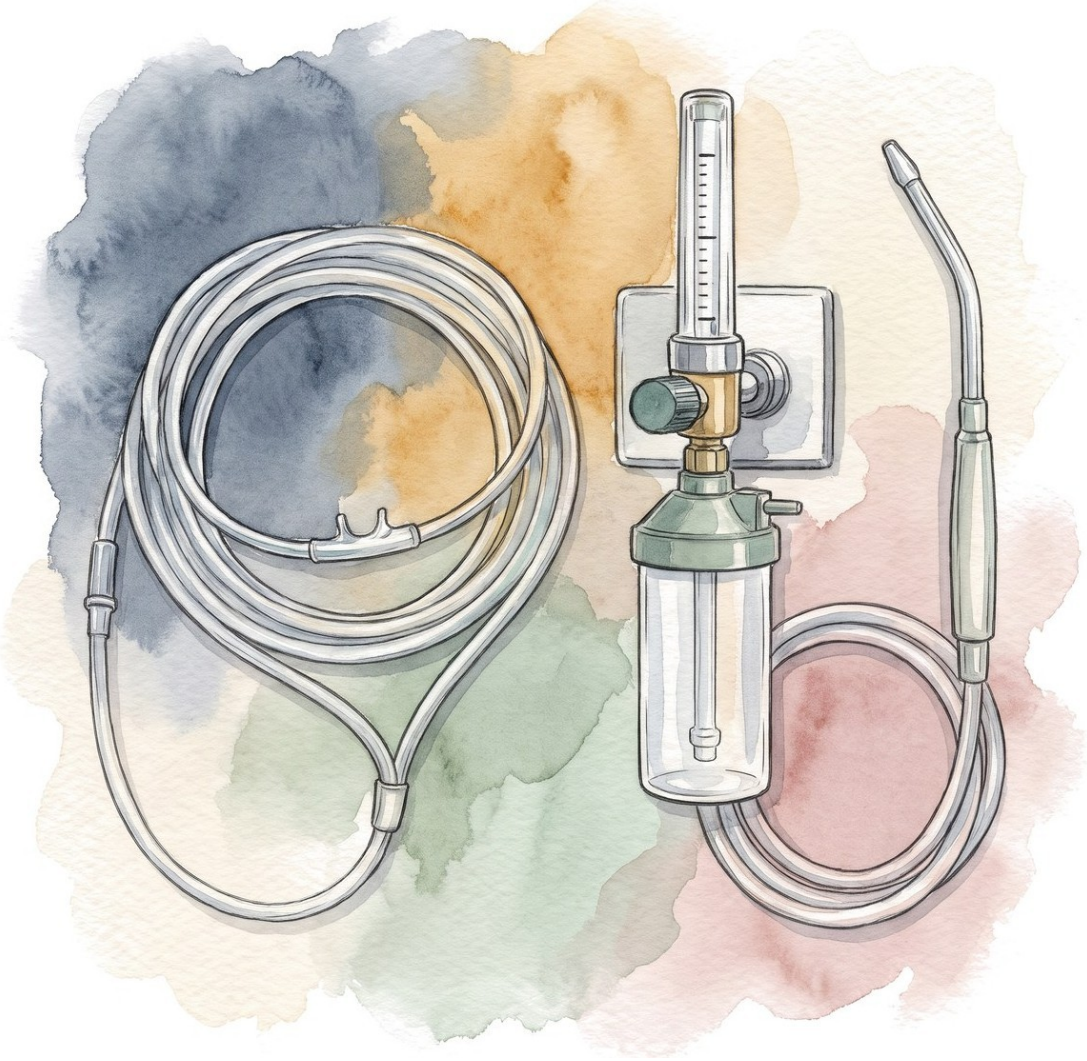
**Not for her**

*AASLD position paper 2011, rec. 26; Alba L, Hay JE, Lee WM et al. J Hepatol 2002;36(Suppl 1):33 (abstract); ACG ALF 2023 [PMID: 37377263]*

# When do you intubate?

- Responds to voice, drifts off, intermittently agitated
- Handling secretions for now
- SpO2 96% on 2 L

**Intubate now, or wait until she can't protect her airway?**



# When do you intubate?

## Grade III-IV encephalopathy in ALF: intubate

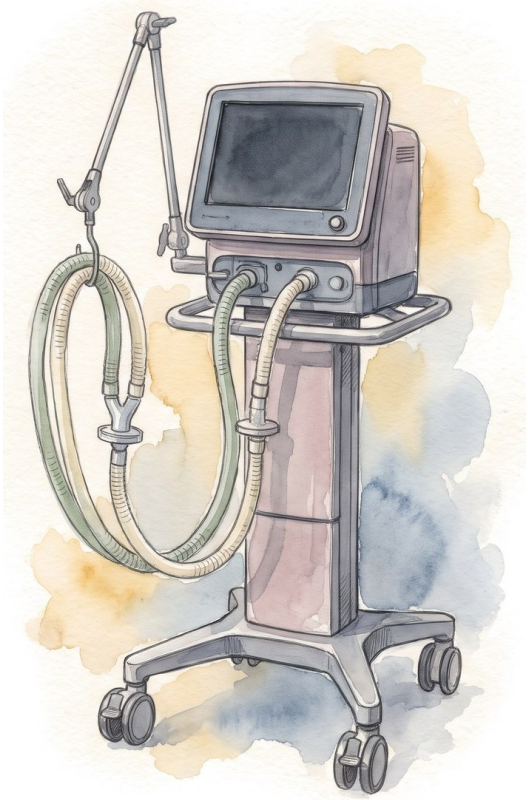
| Before | During | After |
|---|---|---|
| Preoxygenate<br>Norepinephrine ready<br>Check glucose<br>Head of bed 30°, neck neutral | Avoid hypotension (CPP)<br>No coughing or bucking (ICP)<br>Most experienced operator | Propofol: short-acting, may lower ICP<br>Avoid long-acting benzodiazepines<br>Check pupils: sedation hides the exam |

*Practical update 2024 [PMID: 39170946]; ACG ALF 2023 [PMID: 37377263]; SCCM 2020 [PMID: 32058387]*

# Can we use permissive hypercapnia?

*After intubation: bilateral infiltrates, P/F 150 on PEEP 10*

## **Lung protection or brain protection?**



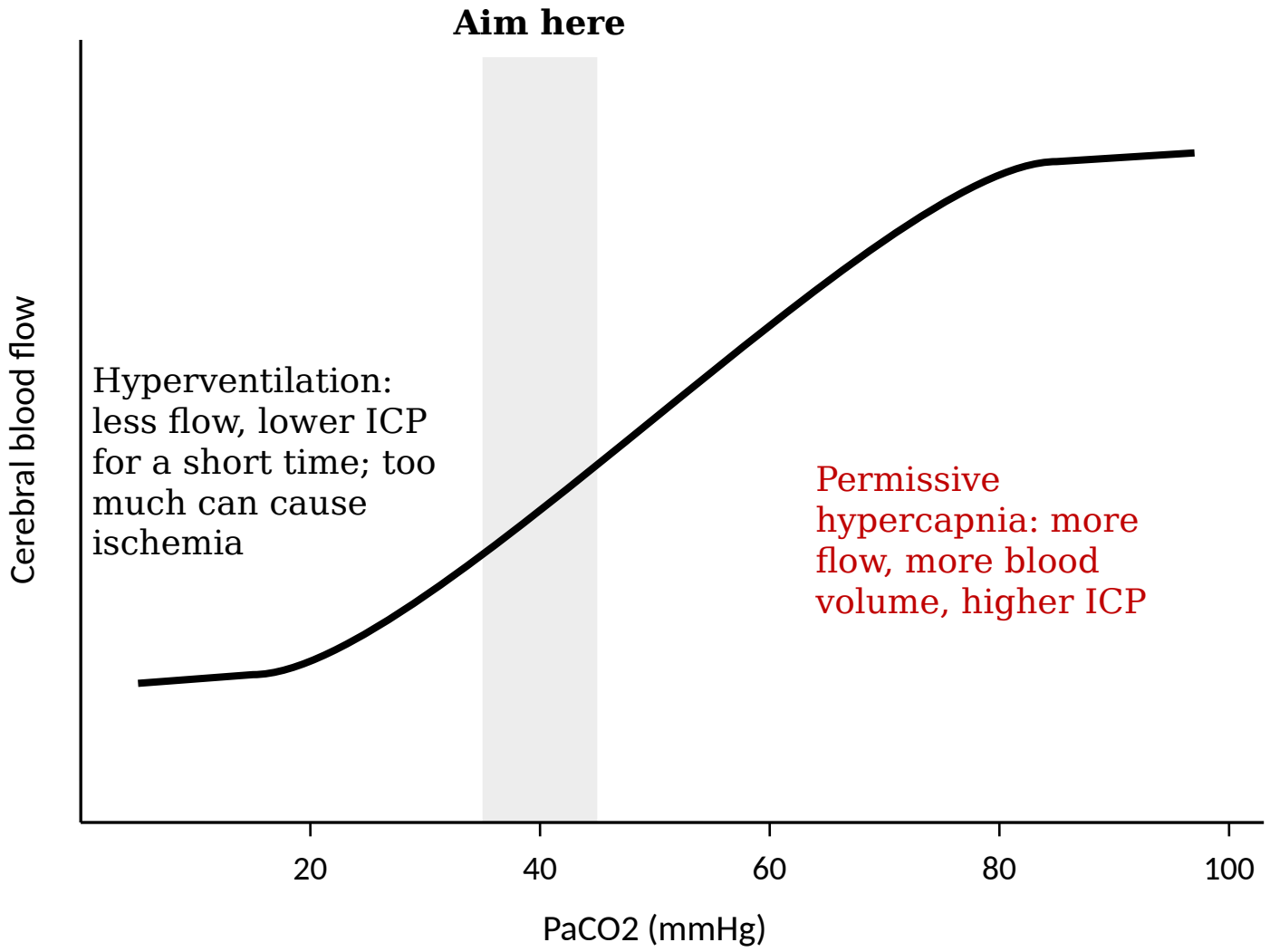
# Can we use permissive hypercapnia?

*After intubation: bilateral infiltrates, P/F 150 on PEEP 10*

| | The lung wants | The brain wants | What we do |
|---|---|---|---|
| **Tidal volume** | 6 mL/kg PBW | Watch the CO2 | 6 mL/kg, raise the rate |
| **CO2** | Tolerate a high CO2 | High CO2 raises ICP | **Keep the CO2 normal** |
| **Hyperventilation** | No | Lowers ICP briefly | Short rescue for herniation only |
| **PEEP** | Recruit | High PEEP: less venous return | Set for O2 and BP; PEEP is fine |

*SCCM 2020 [PMID: 32058387]; practical update 2024 [PMID: 39170946]*

# CO2 and cerebral blood flow

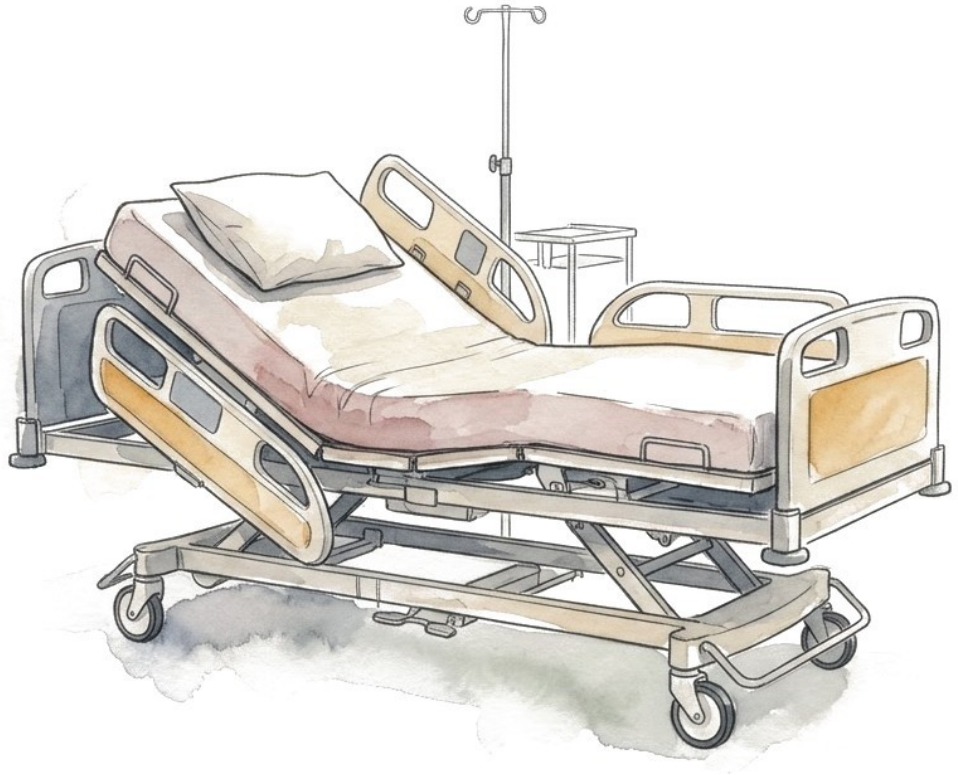



- Between about 20 and 80 mmHg, cerebral blood flow rises with PaCO2
- In a swollen brain, more blood volume means a higher ICP
- Keep PaCO2 normal. Hyperventilate only as a short bridge for herniation

*Schematic, not to scale. Standard cerebral physiology*

# Protecting the brain: every patient with grade III-IV

- Head of bed 30°, neck neutral
- Treat fever, avoid hypoglycemia
- Sodium **145 to 155 mmol/L** with hypertonic saline (use your center's protocol)
- Avoid fluid overload
- Treat seizures, EEG if in doubt. No routine prophylaxis
- No cooling: 34 °C did not prevent intracranial hypertension in an RCT



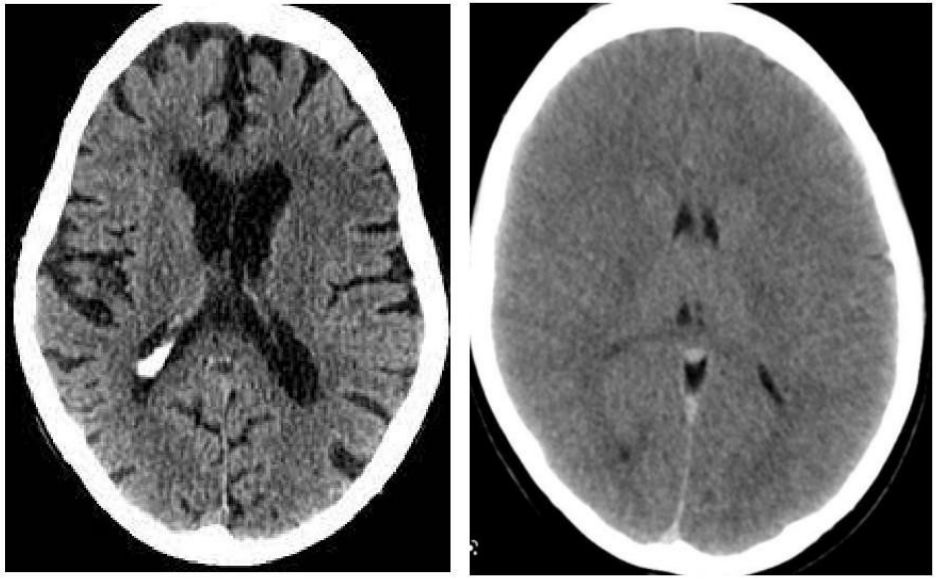
*SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]; Bernal W et al. J Hepatol 2016;65:273-9*

# A surge: new hypertension, bradycardia or a pupil

- Hyperosmolar bolus (hypertonic saline or mannitol)
- Brief hyperventilation as a bridge
- Call the transplant team now

**Things to know**

- Hypertension and bradycardia come late. Don't wait for them
- A normal head CT does not exclude raised ICP
- ICP monitors are not routine; SCCM suggests against (very low certainty)



Normal | Effacement of sulci

*SCCM 2023 [PMID: 37052436]; practical update 2024 [PMID: 39170946]*

# The creatinine is only 1.3

*Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h*

**Do you wait for a conventional dialysis indication?**

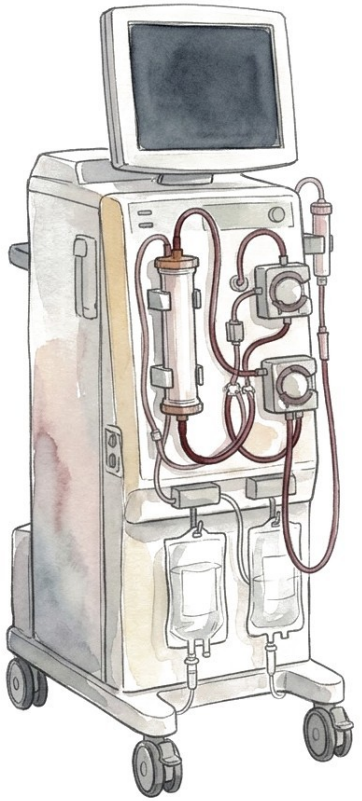


# The creatinine is only 1.3

*Ammonia 182 → 214 µmol/L, grade III, urine output 15 mL/h*

**No. In ALF, start CRRT for the ammonia**

- ACG 2023: suggests early CRRT for hyperammonemia with grade II or higher encephalopathy, even without a renal indication (conditional, very low certainty)
- Continuous, not intermittent: fewer swings in blood pressure and brain water
- Call nephrology now



*ACG ALF guideline 2023 [PMID: 37377263]; update 2022 [PMID: 36001996]*

# Should we correct this INR?

*Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter*

**2 units of FFP first?**



# Should we correct this INR?

*Hour 12: INR 4.1, platelets 88,000. She needs a dialysis catheter*

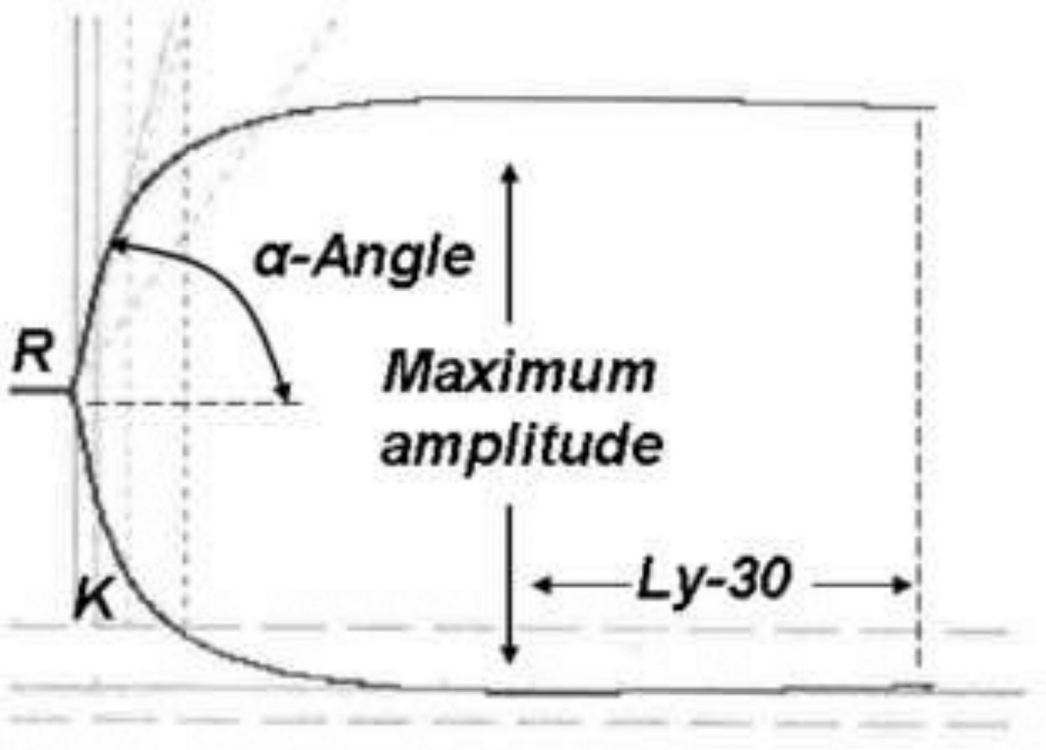
- **No plasma just to fix the number**
- The INR measures only part of clotting, and in ALF it doesn't predict who bleeds
- Plasma adds volume (lungs, brain) and erases the INR trend that prognosis and listing rely on

**What to do instead**

- Ultrasound-guided line, experienced operator, compressible site
- Vitamin K once; check fibrinogen and platelets; viscoelastic testing if available
- Treat bleeding. Plan high-risk procedures (like an ICP monitor) separately

*Update 2022 [PMID: 36001996]; SCCM 2020 [PMID: 32058387]*

# Is her blood thin?

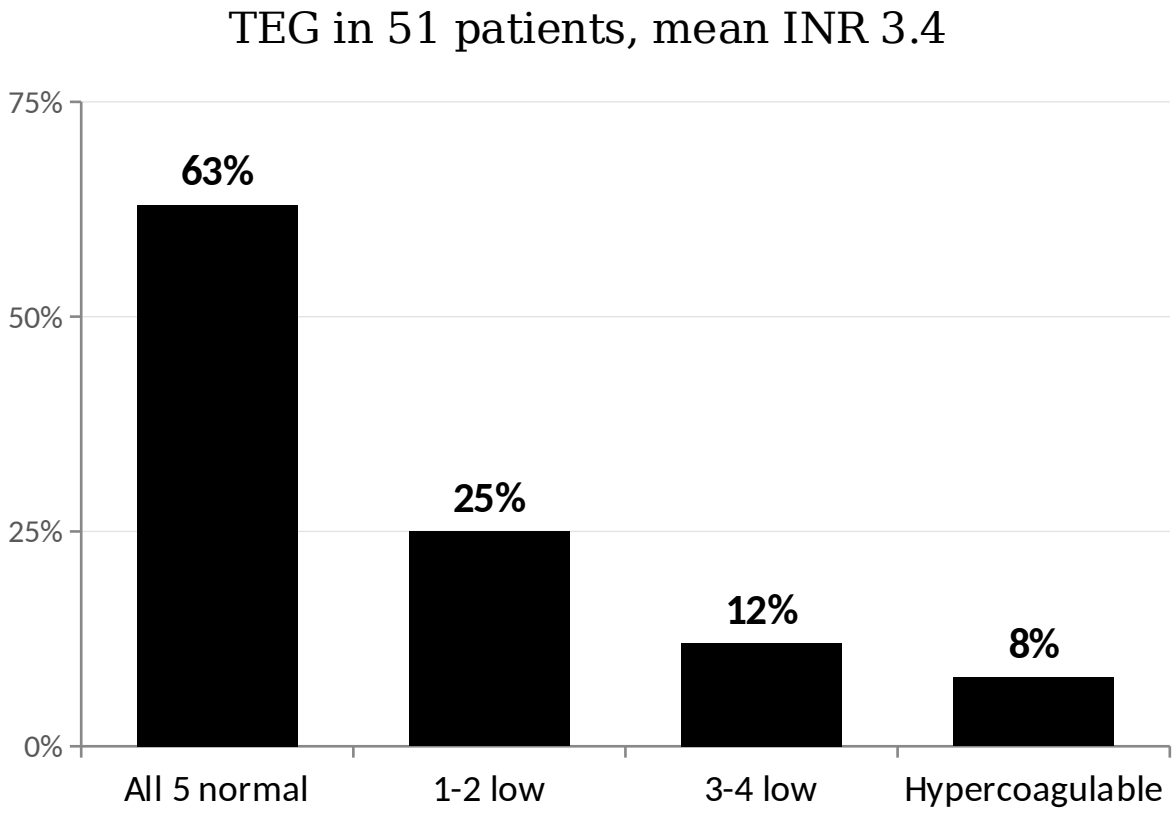



**What a TEG measures in whole blood**

- **R:** time until the clot starts (clotting factors)
- **K and α-angle:** how fast the clot builds (fibrinogen)
- **MA:** how strong the clot gets (platelets, fibrinogen)
- **Ly30:** how much of the clot breaks down in 30 minutes

**Her INR is 4.1. What will her TEG show?**

# Is her blood thin?



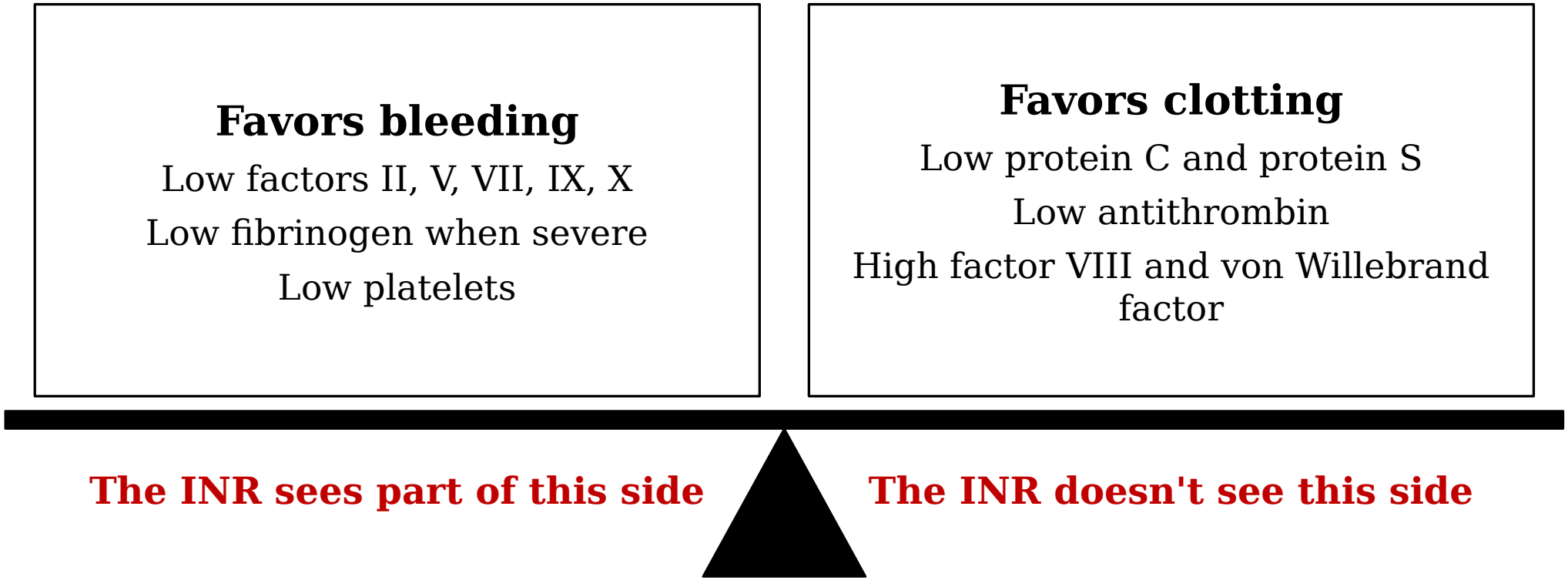

**Stravitz 2012: acute liver injury or ALF, TEG on admission**

- Most had a completely normal TEG, even with INRs up to 9.6
- A weak clot (low MA) only when platelets were below about 126,000
- Patients with encephalopathy had stronger clots than those without
- Factors V and VII were low, but protein C, protein S and antithrombin fell too, and factor VIII was about 4 times normal

**A high INR doesn't mean she will bleed**

*Stravitz RT et al. J Hepatol 2012;56:129-36 [PMID: 21703173]*

# Rebalanced hemostasis




*Balanced in most patients, but easily tipped either way: both bleeding and thrombosis happen*

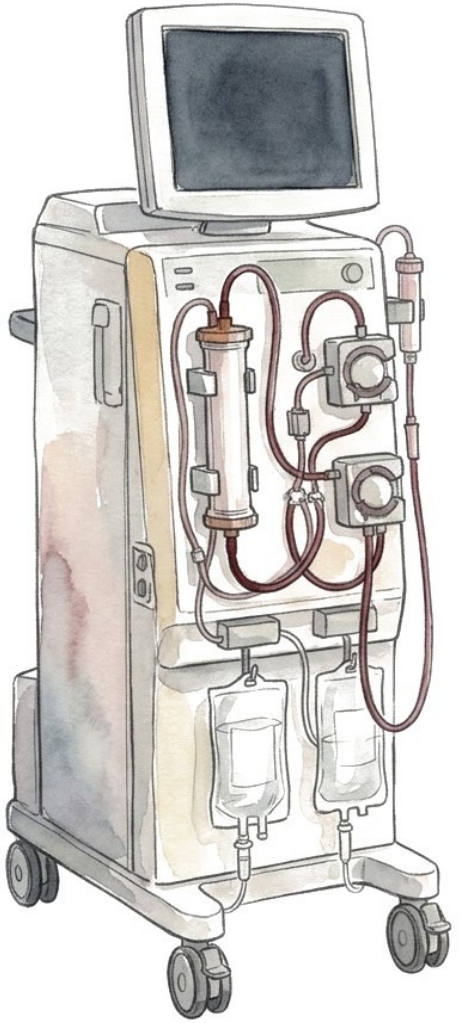
*Concept from Lisman T, Porte RJ. Blood 2010*

# Calling nephrology

**What do you ask them for?**

# Calling nephrology

- It's for the ammonia and the brain, not the creatinine
- Start now and keep it running. A clotted filter means the ammonia climbs back
- One sodium plan: the CRRT fluid and the hypertonic saline together
- Watch phosphate and potassium, and calcium if they use citrate



*ACG ALF 2023 [PMID: 37377263]; update 2022 [PMID: 36001996]*

*I know what you're thinking...*

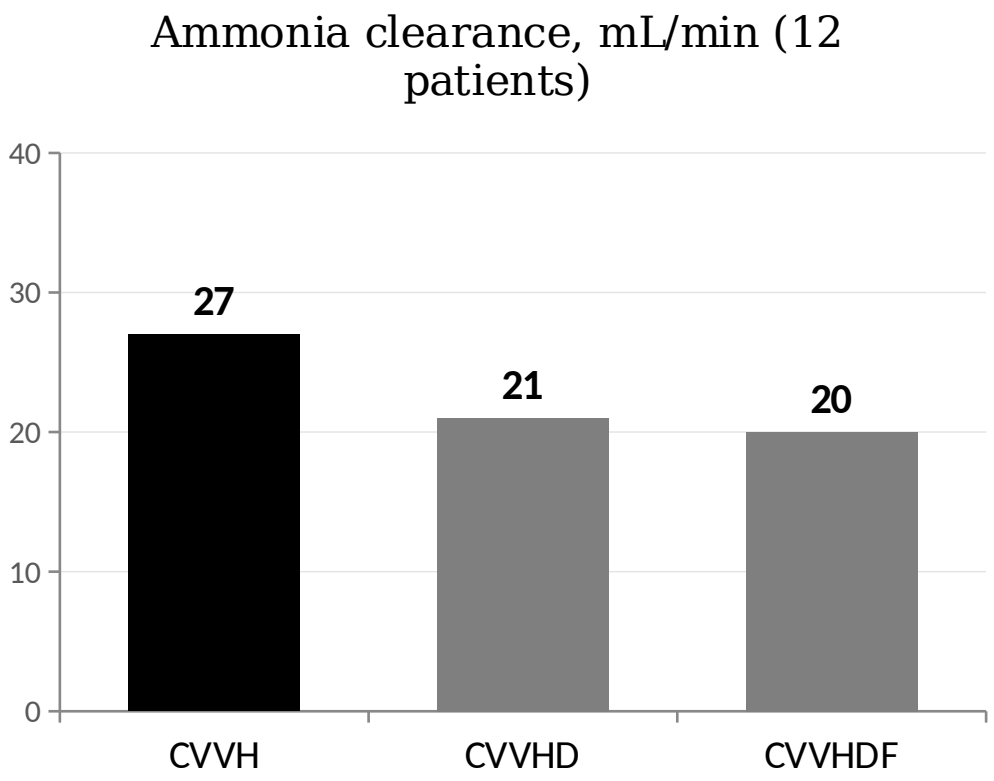
"CVVH doesn't clear ammonia"

**True or false?**


FALSE

# "CVVH doesn't clear ammonia"

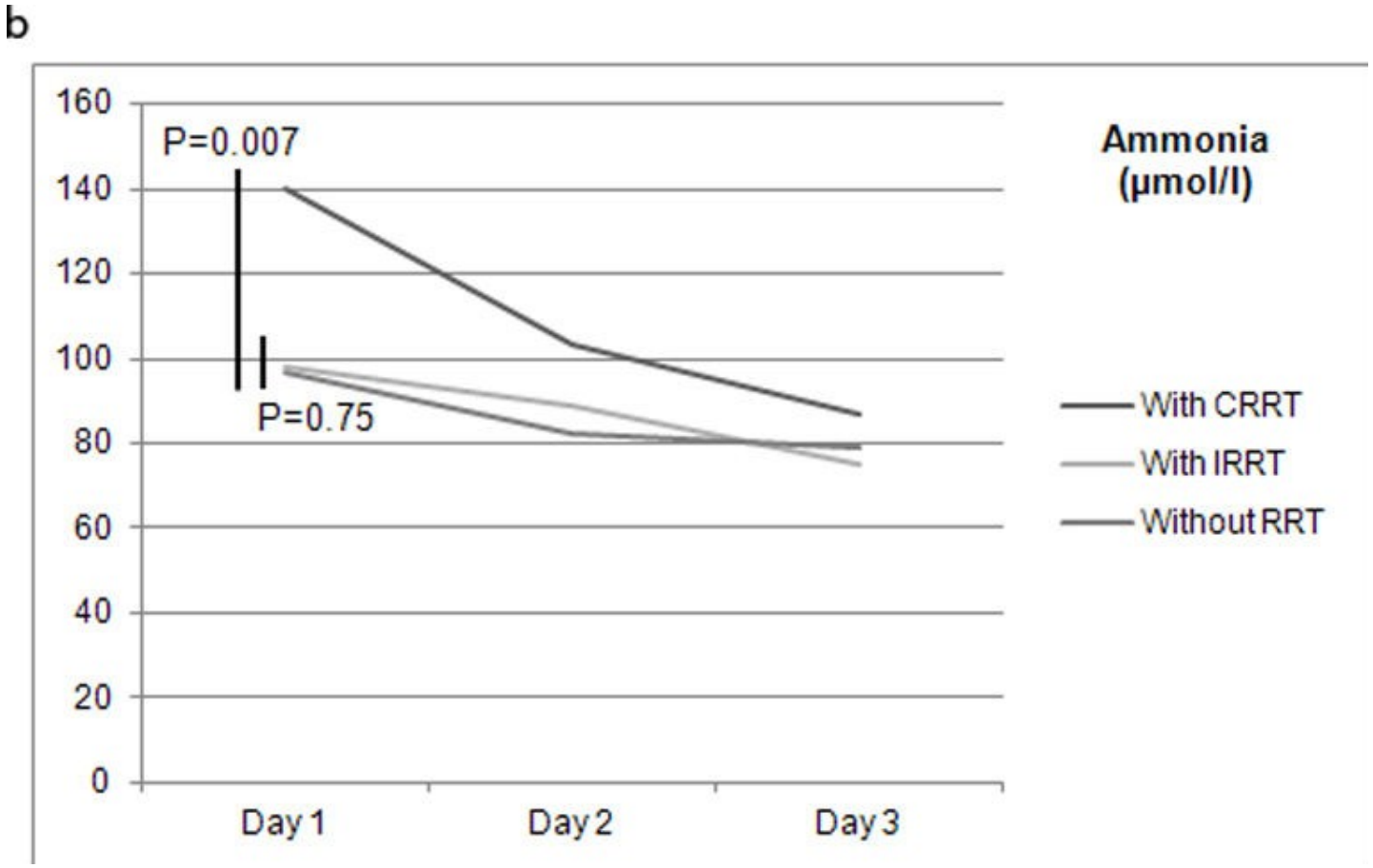
- Ammonia is tiny (17 Da), water soluble and not protein bound
- It crosses the filter with the water (convection, CVVH) or down its gradient (diffusion, CVVHD)
- So what matters is effluent flow and hours on a working circuit, not the letters
- More effluent, more clearance: **90 vs 35 mL/kg/h** (Slack 2014)
- 54 ALF patients on CRRT: median ammonia **151 → 52 µmol/L** by day 5 (Warrillow 2020)




**It does. About half the clearance of urea, so dose and filter life matter**

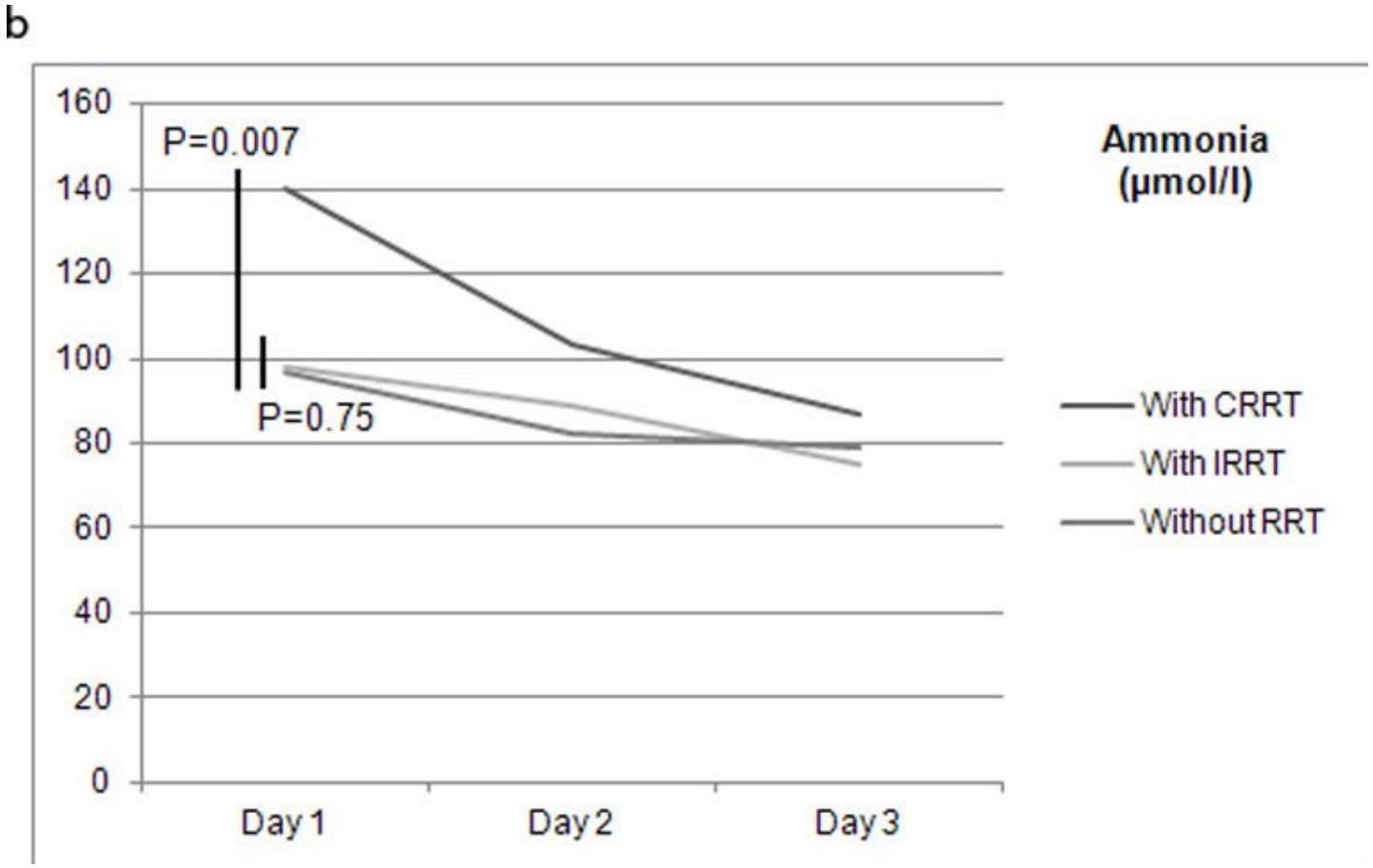
*Fisher C et al. Blood Purif 2022 [PMID: 35042216]; Warrillow SJ et al. Crit Care Med 2020 [PMID: 31939790]; Slack AJ et al. Liver Int 2014*

# Does early CRRT actually help?




*Cardoso FS et al. Hepatology 2018;67:711-20 [PMID: 28859230]*

# Does early CRRT actually help?




*Ammonia fell from about 140 to under 90 µmol/L over 3 days on CRRT. Little change with intermittent RRT or no RRT*

*Cardoso FS et al. Hepatology 2018;67:711-20 [PMID: 28859230]*

# Does early CRRT actually help?

| | US ALF Study Group (2018) | Pilot RCT (2026) |
|---|---|---|
| Design | Registry, observational | Open label, ALF with cerebral edema |
| Comparison | CRRT vs no CRRT | CRRT within 12 h vs delayed |
| Finding | | |
| Catch | | |

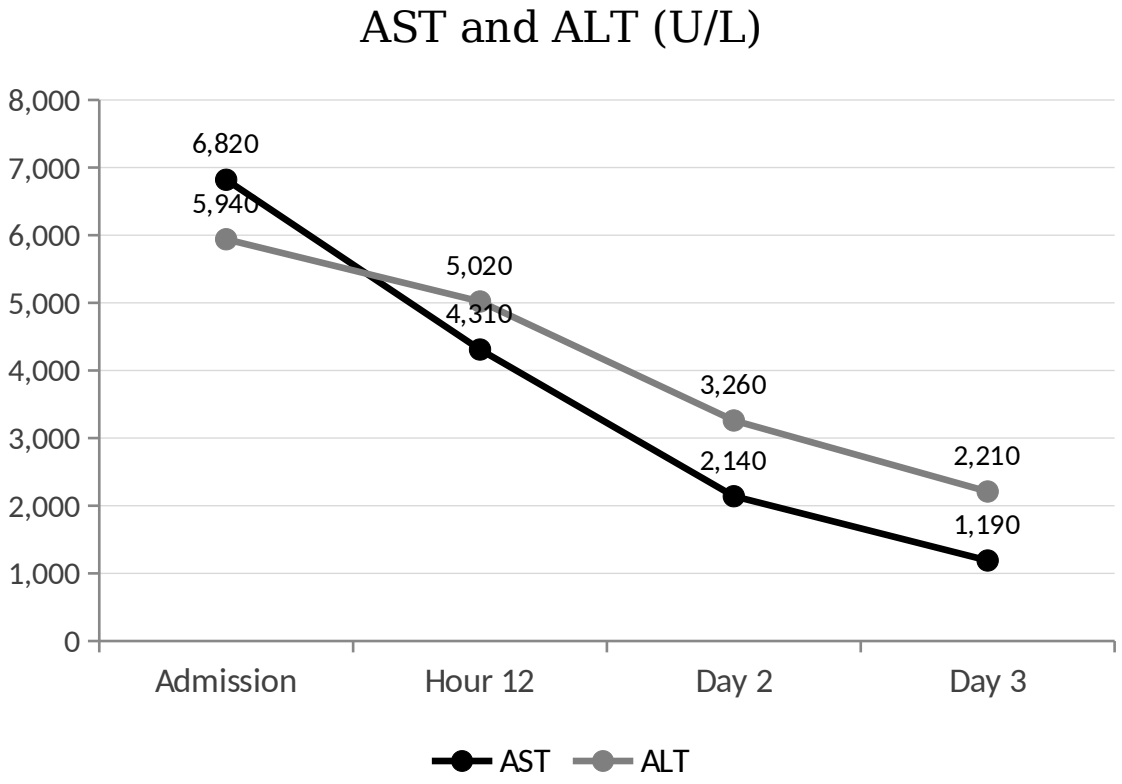
*Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]*

# Does early CRRT actually help?

| | US ALF Study Group (2018) | Pilot RCT (2026) |
|---|---|---|
| Design | Registry, observational | Open label, ALF with cerebral edema |
| Comparison | CRRT vs no CRRT | CRRT within 12 h vs delayed |
| Finding | Lower ammonia, better survival | Same 28-day survival; fewer deaths by day 7 |
| Catch | **Association; selection bias** | **Small, one center, protocol violations** |

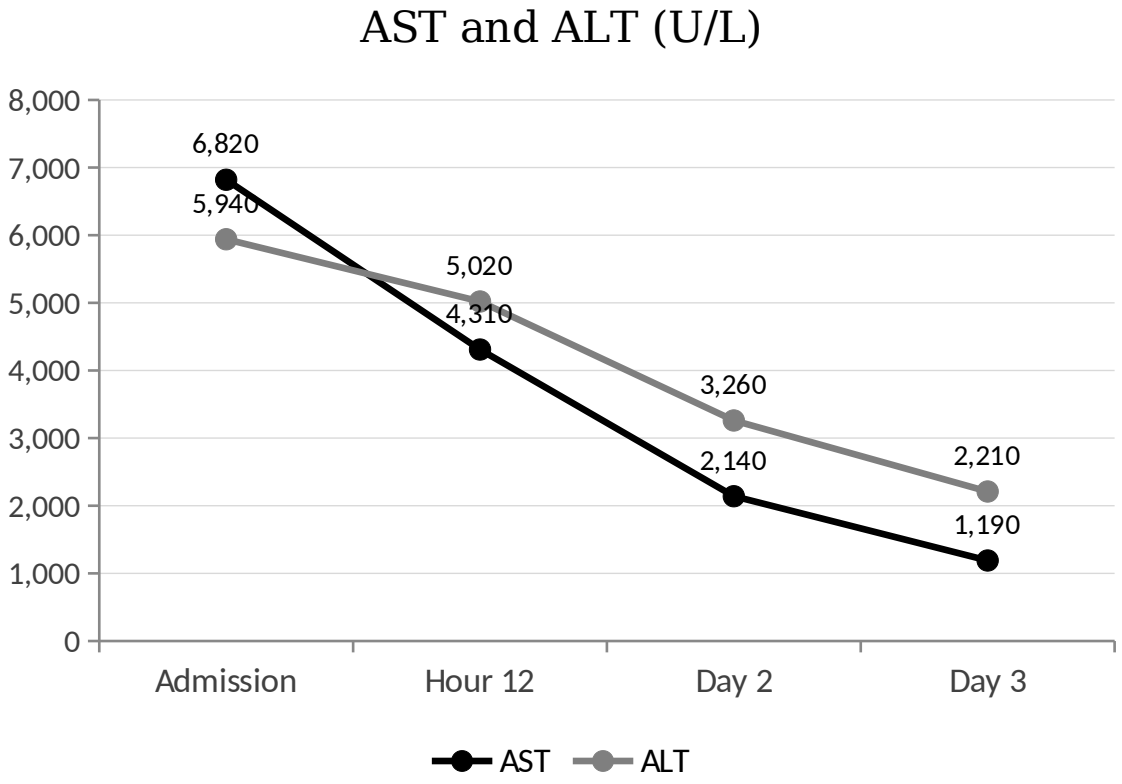
*Cardoso FS et al. Hepatology 2018 [PMID: 28859230]; Aliment Pharmacol Ther 2026 [PMID: 42332991]*

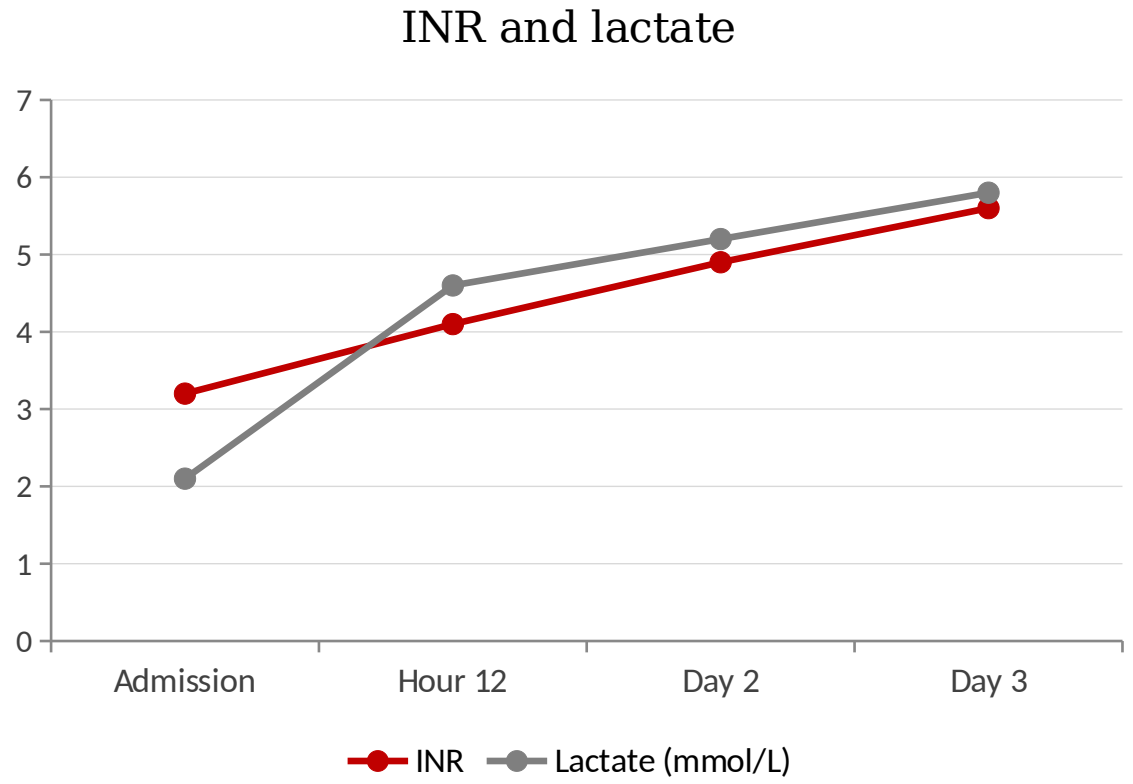
# Day 3: the enzymes are falling




**Is she getting better?**

# Day 3: the enzymes are falling




*Day 2 arterial pH 7.26 after fluids*

**No. The INR, the lactate and her brain are all worse**

*Made-up numbers for teaching*

# When do you call transplant hepatology?

**When she meets King's College criteria?**

**When MELD is high enough?**



# When do you call transplant hepatology?

**You should have called on admission, when you first suspected ALF**

- King's College criteria predict death without transplant. They are specific, not sensitive
- They are not a referral threshold. If you wait for them, the evaluation starts too late
- Patient A: arterial pH **7.26** on day 2 after fluids. She now meets the acetaminophen criteria



*O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882]; full criteria in the appendix*

# Can plasma exchange buy time?

Research Article

CrossMark

EASL JOURNAL OF HEPATOLOGY

**High-volume plasma exchange in patients with acute liver failure: An open randomised controlled trial**

Fin Stolze Larsen[1,*], Lars Ebbe Schmidt[1], Christine Bernsmeier[2], Allan Rasmussen[3], Helena Isoniemi[4], Vishal C. Patel[2], Evangelos Triantafyllou[2], William Bernal[2], Georg Auzinger[2], Debbie Shawcross[2], Martin Eefsen[1], Peter Nissen Bjerring[1], Jens Otto Clemmesen[1], Krister Hockerstedt[4], Hans-Jørgen Frederiksen[5], Bent Adel Hansen[1], Charalambos G. Antoniades[2,6,†], Julia Wendon[2,†]

## What do you think they found?

*Larsen FS et al. J Hepatol 2016;64:69-78 [PMID: 26325537]*

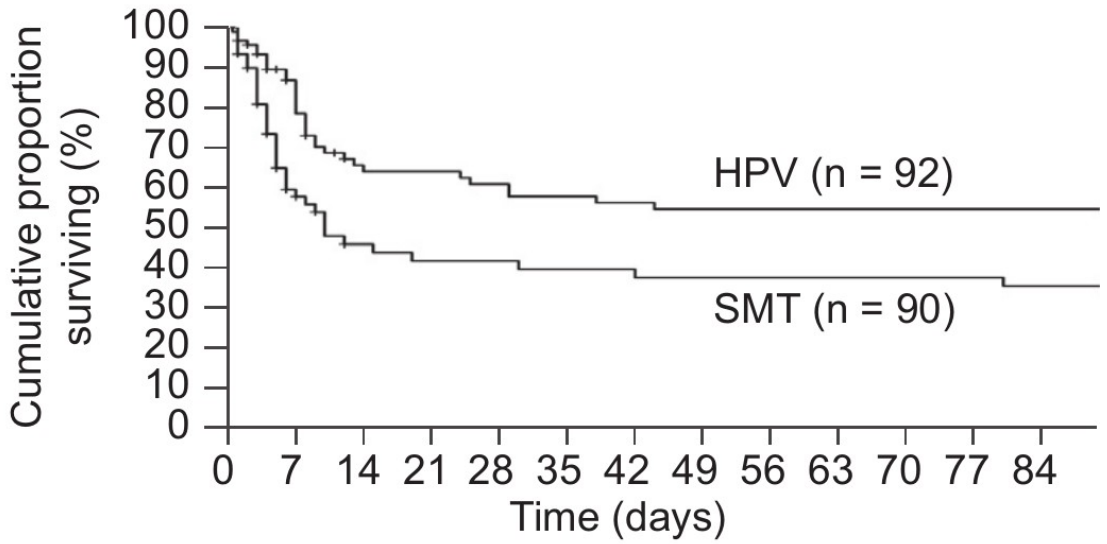
# Can plasma exchange buy time?

Research Article

EASL JOURNAL OF HEPATOLOGY

## High-volume plasma exchange in patients with acute liver failure: An open randomised controlled trial

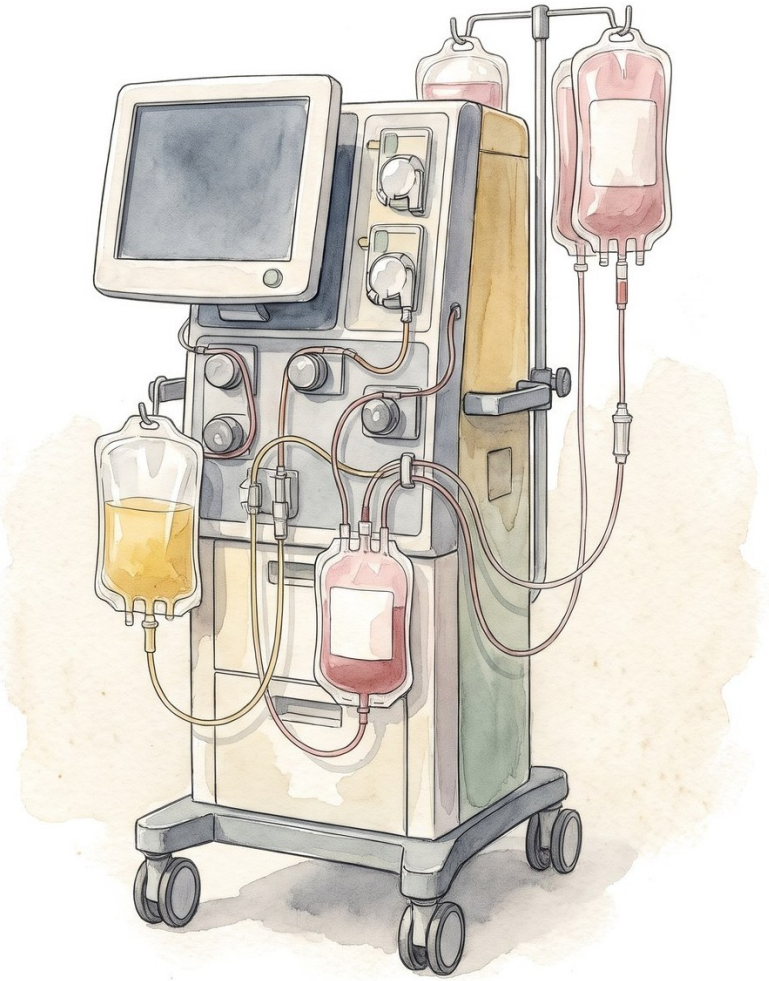
Fin Stolze Larsen[1,*], Lars Ebbe Schmidt[1], Christine Bernsmeier[2], Allan Rasmussen[3], Helena Isoniemi[4], Vishal C. Patel[2], Evangelos Triantafyllou[2], William Bernal[2], Georg Auzinger[2], Debbie Shawcross[2], Martin Eefsen[1], Peter Nissen Bjerring[1], Jens Otto Clemmesen[1], Krister Hockerstedt[4], Hans-Jørgen Frederiksen[5], Bent Adel Hansen[1], Charalambos G. Antoniades[2,6,†], Julia Wendon[2,†]




*Larsen FS et al. J Hepatol 2016;64:69-78 [PMID: 26325537]*

# Can plasma exchange buy time?

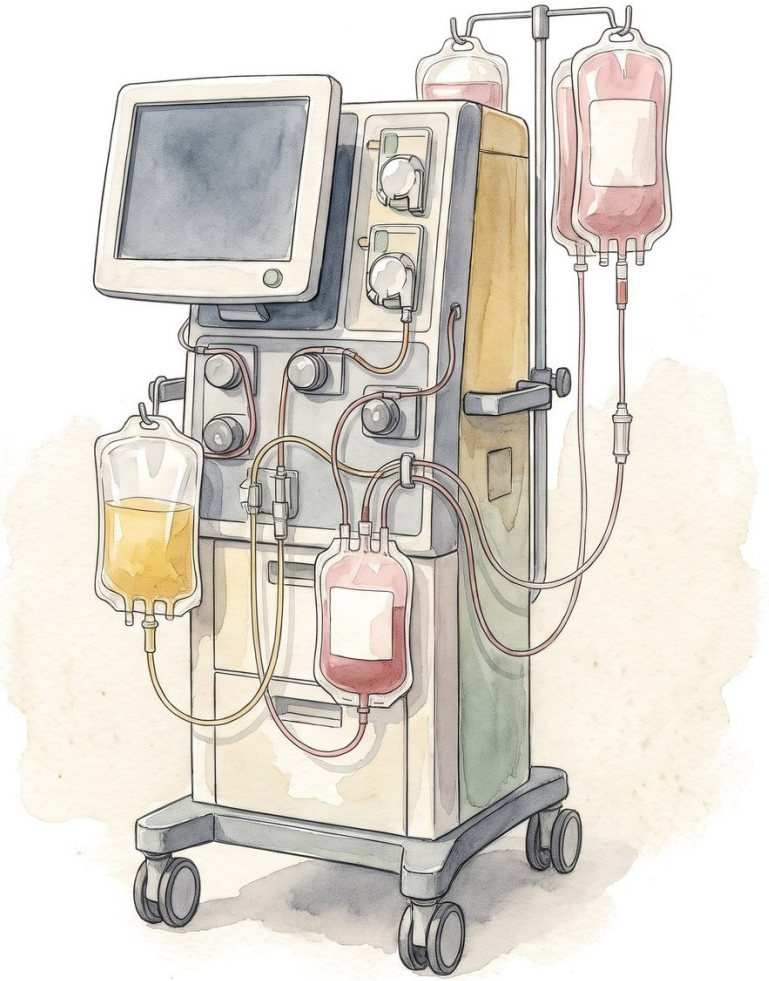
| | |
|---|---|
| **Study** | RCT, 182 ALF: high-volume plasma exchange × 3 days |
| **Finding** | Hospital survival 58.7% vs 47.8% (HR 0.56) |
| **Limits** | |
| **So** | |



*Larsen FS et al. J Hepatol 2016 [PMID: 26325537]*

# Can plasma exchange buy time?

| | |
|---|---|
| **Study** | RCT, 182 ALF: high-volume plasma exchange × 3 days |
| **Finding** | Hospital survival 58.7% vs 47.8% (HR 0.56) |
| **Limits** | **Not blinded; nothing on ACLF** |
| **So** | A bridge; don't delay listing |



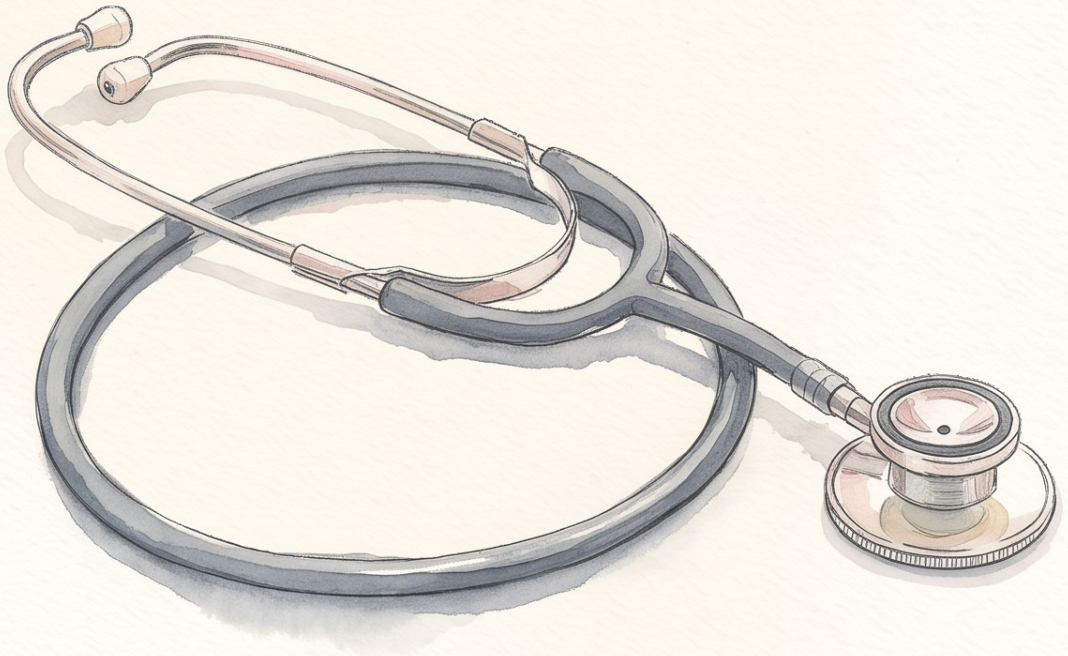
*Larsen FS et al. J Hepatol 2016 [PMID: 26325537]*

# Let's recap: bed 4, day 3

| Problem | What she's on | What we watch |
|---|---|---|
| Acetaminophen ALF, pH 7.26 | NAC; listed 1A on day 2 | Time to an offer |
| HE III-IV, ammonia 182 | Intubated, CO2 normal, Na 145-155, CRRT | Pupils, ammonia, Na |
| Lactate 5.8 and rising | Volume, MAP, look for infection | Lactate, MAP, pressors |
| INR 5.6, not bleeding | **No plasma** | Bleeding, INR trend |
| Infection risk | Surveillance cultures | Low bar for antimicrobials |

*Made-up numbers for teaching*

# Any questions so far?



# Bed 7

56M, cirrhosis and fever



# Back to day 1: bed 7

| Bed 4 | Bed 7 |
|---|---|
| 28-year-old woman | 56-year-old man |
| Acetaminophen ALF | Alcohol-related cirrhosis with ascites |
| Intubated, on CRRT | Fever, confusion, low blood pressure |

**Is this liver failure too?**

# 56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
- Urine output 15 mL/h

| | Now | Baseline |
|---|---|---|
| Bilirubin (mg/dL) | 5.8 | 2.1 |
| INR | 2.1 | 1.6 |
| Creatinine (mg/dL) | 2.6 | 0.9 |
| Sodium (mmol/L) | 127 | 133 |
| WBC (/µL) | 16,200 | |
| Platelets (/µL) | 62,000 | |
| Lactate (mmol/L) | 3.9 | |

**What's your next step?**

# 56M, cirrhosis and fever

- Alcohol-related cirrhosis, ascites on diuretics; last drink 3 weeks ago
- Temp 38.4 °C, HR 112, BP 82/46
- RR 26, SpO2 92% on a 35% Venturi mask; PaO2 74
- Confused, asterixis (grade II)
- Urine output 15 mL/h

| | Now | Baseline |
|---|---|---|
| Bilirubin (mg/dL) | 5.8 | 2.1 |
| INR | 2.1 | 1.6 |
| Creatinine (mg/dL) | 2.6 | 0.9 |
| Sodium (mmol/L) | 127 | 133 |
| WBC (/µL) | 16,200 | |
| Platelets (/µL) | 62,000 | |
| Lactate (mmol/L) | 3.9 | |

**What's your next step?**

**Find the source: cultures, antibiotics**

What tipped him over?

**INR 2.1, platelets 62,000. Do you tap him?**



# What tipped him over?

- **Yes, now. No plasma, no platelets**
- Ascitic fluid: **PMN 1,240/mm³**. Spontaneous bacterial peritonitis

**Common triggers**

- Infection: SBP, pneumonia, urine, lines, C. difficile
- Alcohol-associated hepatitis
- GI bleeding
- Drugs: NSAIDs, sedatives, too much diuretic
- HBV flare, ischemia
- No trigger is found in some patients. Repeat the search if he gets worse



*EASL ACLF 2023 [PMID: 37364789]; SCCM 2023 [PMID: 37052436]*

# Why is he hypotensive?

| Mechanism | What you do |
|---|---|
| | |
| | |
| | |
| | |
| | |
| | |

*Surviving Sepsis 2026 [PMID: 41869847]; SCCM 2020 [PMID: 32058387]; EASL ACLF 2023 [PMID: 37364789]*

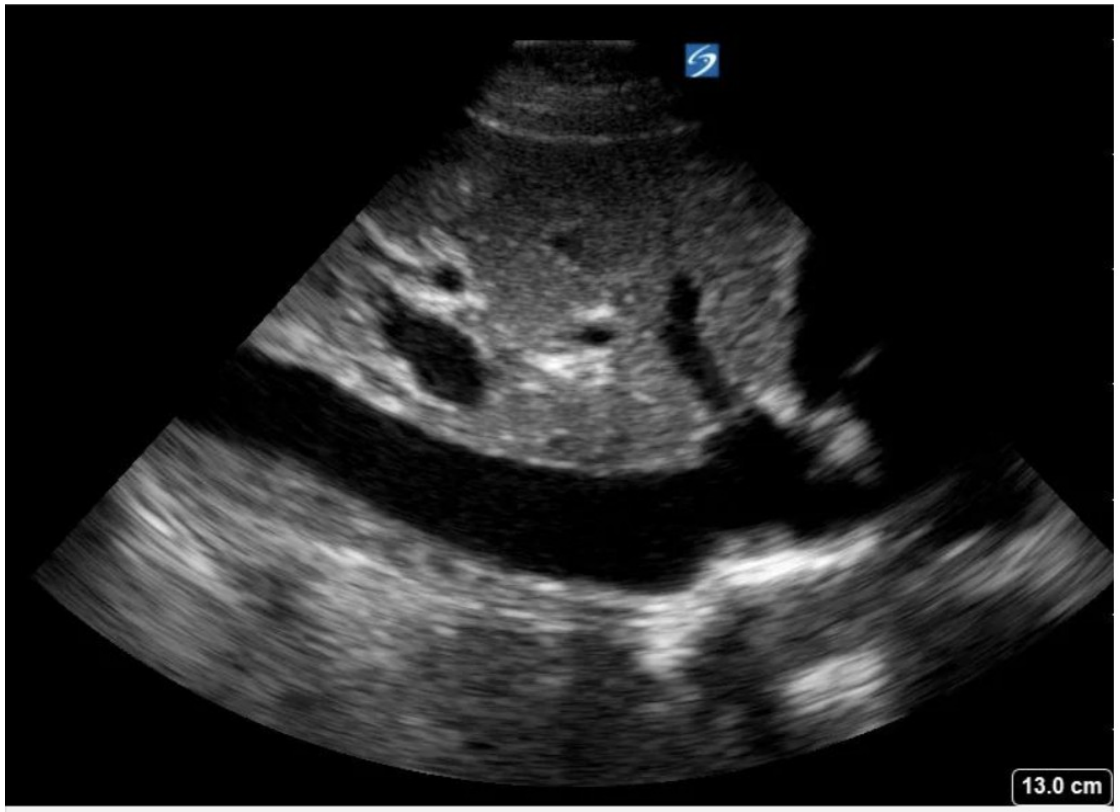
# Why is he hypotensive?

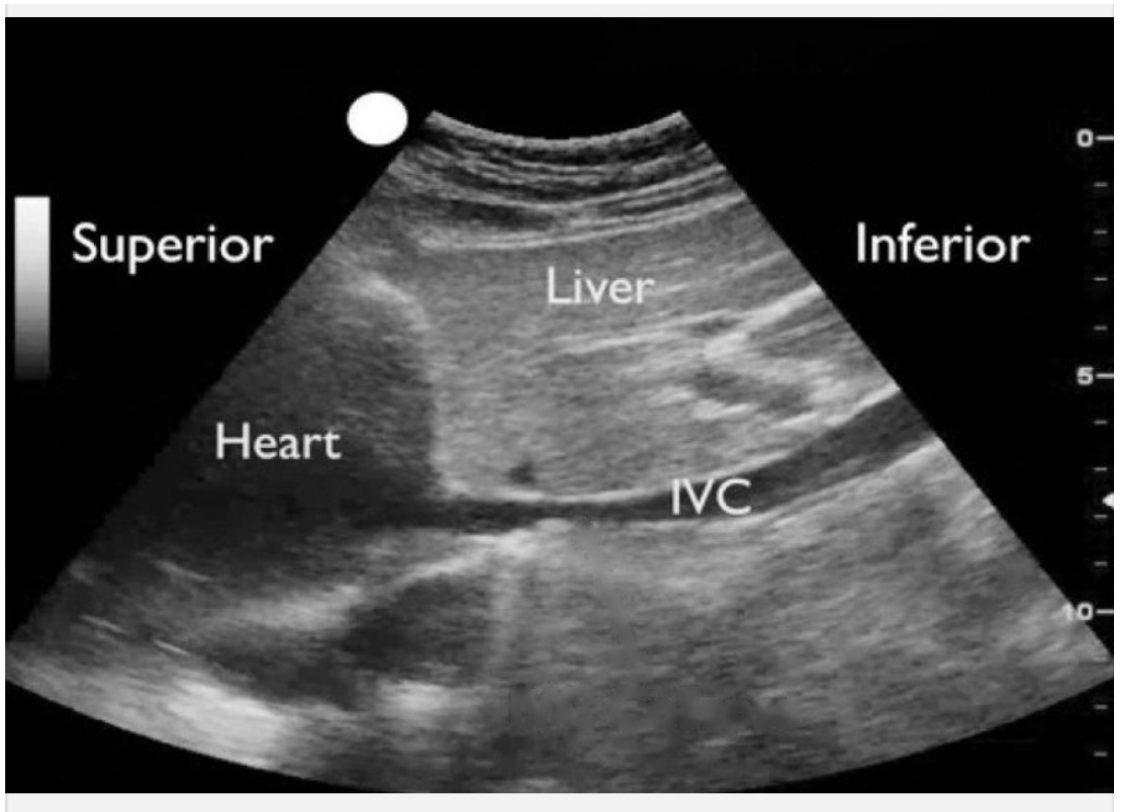
| Mechanism | What you do |
|---|---|
| Sepsis (SBP) | Cultures, antibiotics within 1 h, source control |
| Splanchnic vasodilation | Norepinephrine, MAP 65, adjust to perfusion |
| Too little volume, or too much | Ultrasound and dynamic measures |
| Hidden bleeding | Hemoglobin, rectal exam, think varices |
| Relative adrenal insufficiency | Hydrocortisone if shock is refractory |
| Cirrhotic cardiomyopathy | Echo |

**Usually more than one of these**

*Surviving Sepsis 2026 [PMID: 41869847]; SCCM 2020 [PMID: 32058387]; EASL ACLF 2023 [PMID: 37364789]*

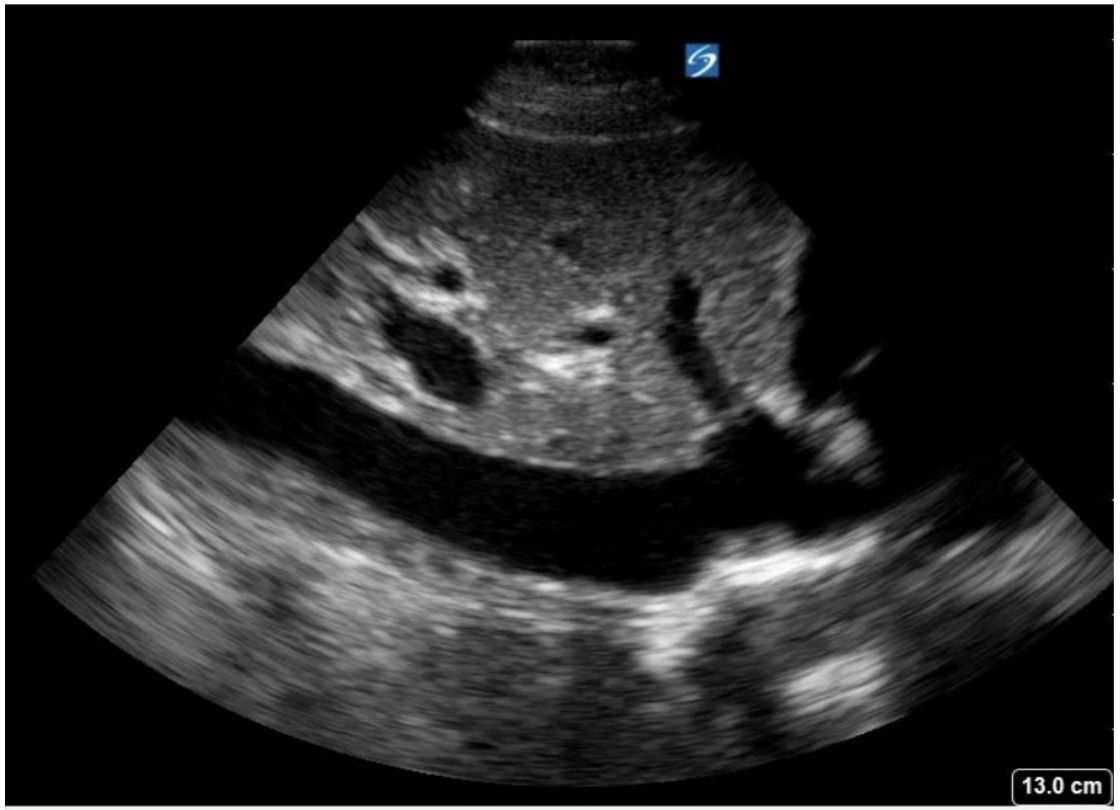
# Too little volume, or too much?

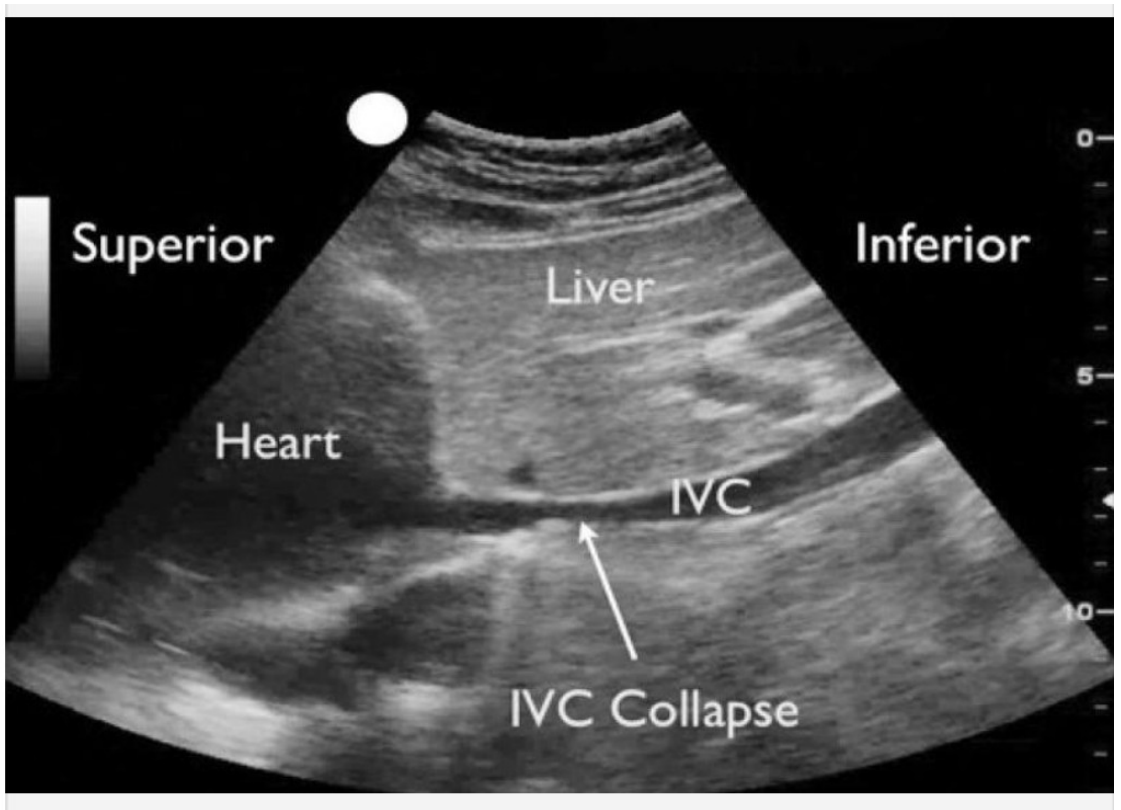



**Which one would you give more fluid?**

# Too little volume, or too much?

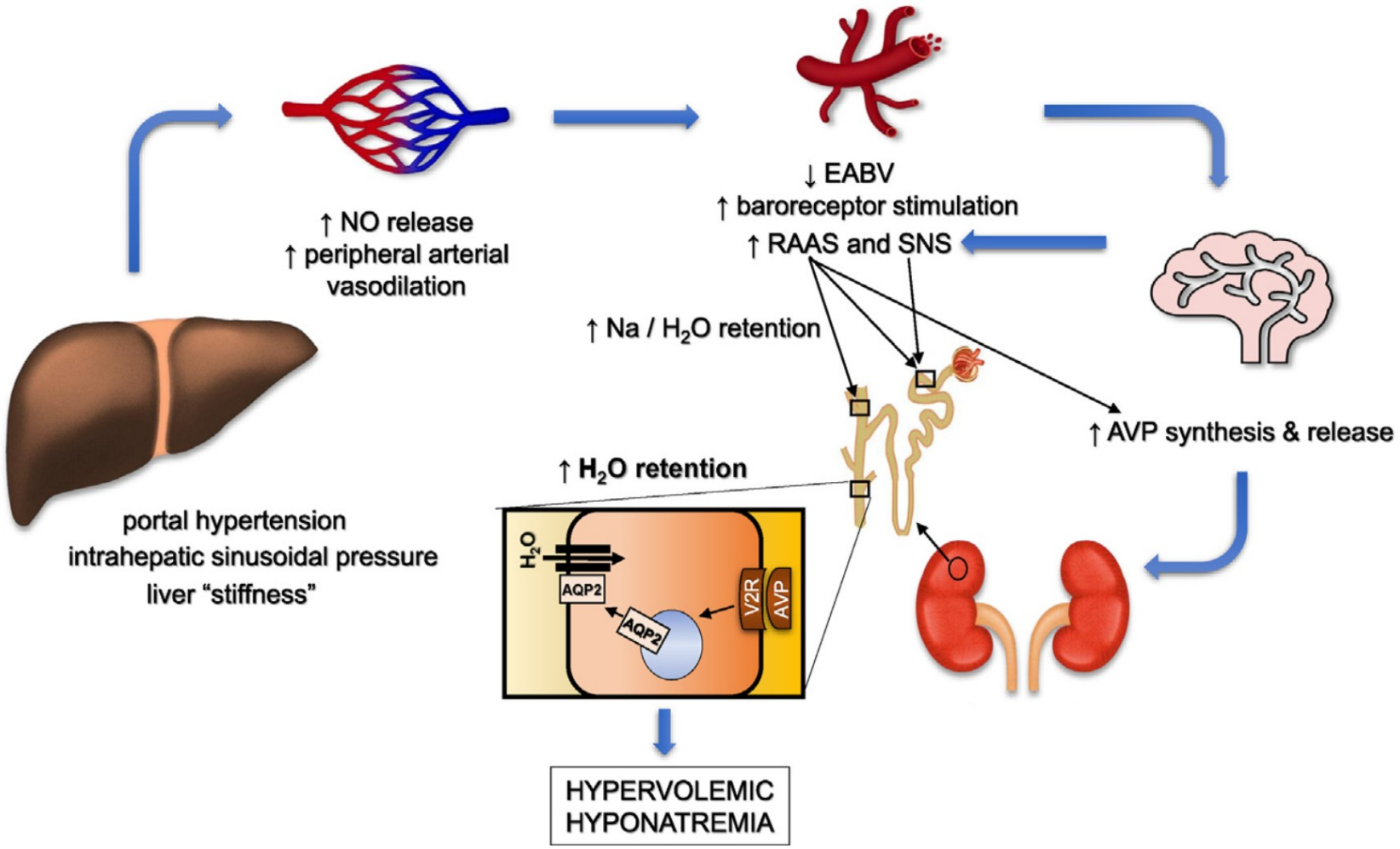



*Left: a full, plethoric IVC. Right: an IVC that collapses*

**Look before you give the next liter**

# Splanchnic vasodilation in cirrhosis




*Rondon-Berrios H, Velez JCQ. Clin Liver Dis 2022*

# ALF, decompensation or ACLF?

| | Acute liver failure | Acute decompensation | ACLF |
|---|---|---|---|
| **Liver before** | | | |
| **Defined by** | | | |
| **Usually kills** | | | |
| **ICU focus** | | | |
| **Transplant** | | | |

*EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]*

# ALF, decompensation or ACLF?

| | Acute liver failure | Acute decompensation | ACLF |
|---|---|---|---|
| **Liver before** | No cirrhosis | Cirrhosis | Cirrhosis or chronic liver disease |
| **Defined by** | INR ≥ 1.5 + HE | Ascites, bleeding, HE, jaundice | Decompensation + organ failure |
| **Usually kills** | Brain, sepsis | Bleeding, infection | Infection, shock, organ failure |
| **ICU focus** | Brain, ammonia, early CRRT | Treat the trigger | Source, shock, kidney vs lung |
| **Transplant** | Urgent listing | Refer by MELD or complication | Narrow window; reassess daily |

*EASL ALF 2017 [PMID: 28417882]; EASL ACLF 2023 [PMID: 37364789]*

# Count the failing organs

| Organ | Failure (EASL-CLIF) |
|---|---|
| Liver | Bilirubin ≥ 12 mg/dL |
| Kidney | Creatinine ≥ 2.0 mg/dL or on RRT |
| Brain | Encephalopathy grade III-IV |
| Coagulation | INR ≥ 2.5 |
| Circulation | Vasopressors (or terlipressin) |
| Lungs | PaO2/FiO2 ≤ 200, SpO2/FiO2 ≤ 214, or ventilated (not for encephalopathy) |

*CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]*

# Count the failing organs

| Organ | Failure (EASL-CLIF) | Patient B |
|---|---|---|
| Liver | Bilirubin ≥ 12 mg/dL | 5.8: no |
| Kidney | Creatinine ≥ 2.0 mg/dL or on RRT | **2.6: yes** |
| Brain | Encephalopathy grade III-IV | Grade II: no |
| Coagulation | INR ≥ 2.5 | 2.1: no |
| Circulation | Vasopressors (or terlipressin) | **Norepinephrine: yes** |
| Lungs | PaO2/FiO2 ≤ 200, SpO2/FiO2 ≤ 214, or ventilated (not for encephalopathy) | PaO2/FiO2 211: no |

**Two organ failures: ACLF grade 2**

*CLIF-C organ failure score, EASL ACLF 2023 [PMID: 37364789]; CANONIC [PMID: 23474284]*

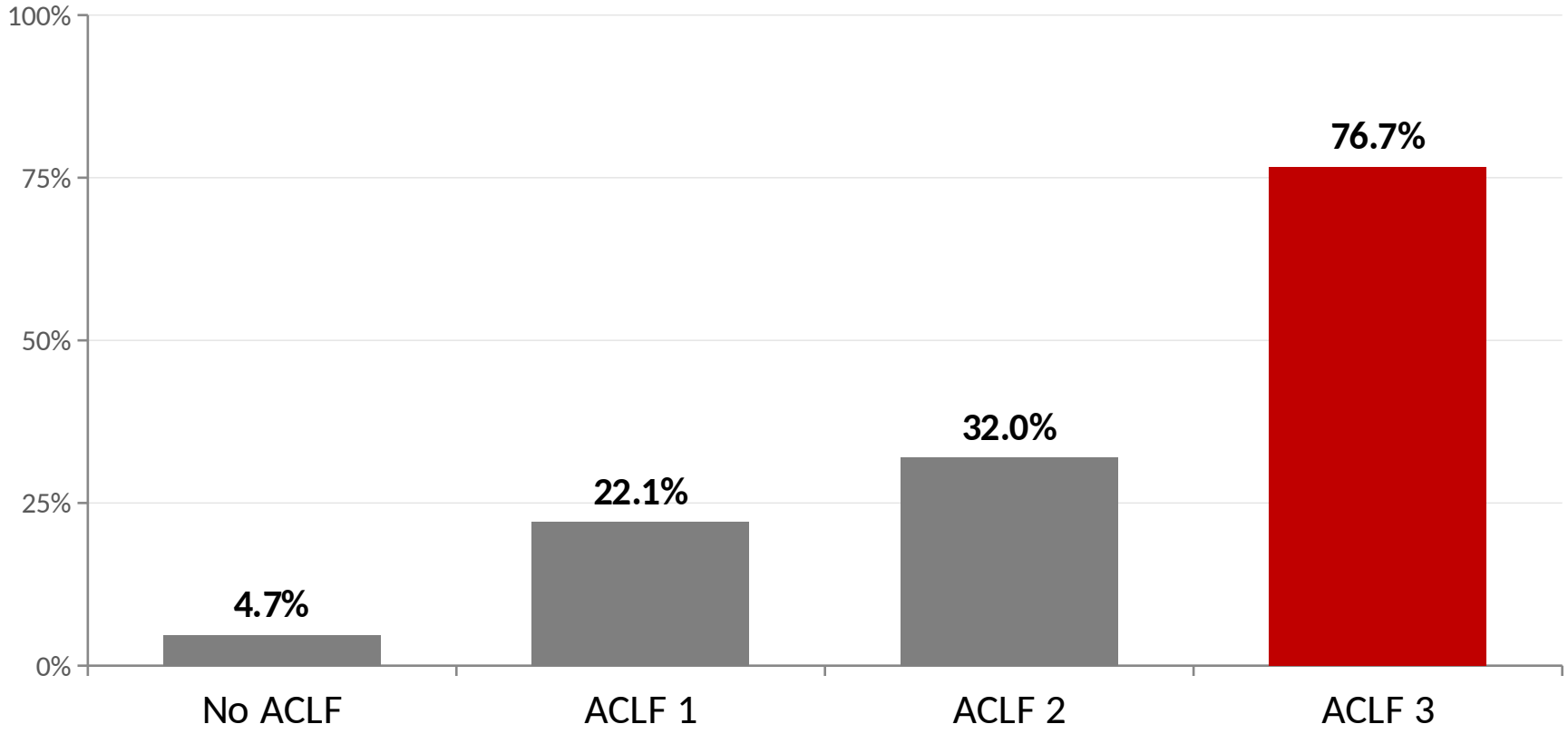
# ACLF grades

| Grade | What it takes |
|---|---|
| No ACLF | No organ failure<br>or one liver, clotting, circulation or lung failure, creatinine < 1.5, no HE<br>or brain failure alone, creatinine < 1.5 |
| ACLF 1 | Kidney failure alone<br>or one liver, clotting, circulation or lung failure + creatinine 1.5-1.9 or HE I-II<br>or brain failure alone + creatinine 1.5-1.9 |
| ACLF 2 | **Two organ failures** |
| ACLF 3 | **Three or more** |

*A patient with "no ACLF" can still be very sick. Use the CLIF-C calculator*

*Moreau R et al. Gastroenterology 2013 [PMID: 23474284]; EASL ACLF 2023 [PMID: 37364789]*

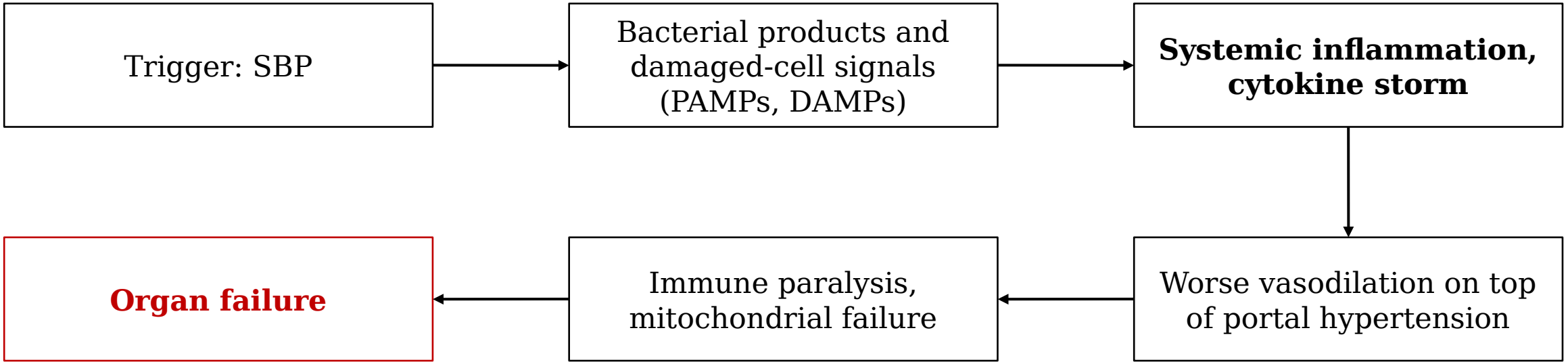
# 28-day mortality by ACLF grade




*CANONIC, 1,343 patients, 2013*

*Moreau R et al. Gastroenterology 2013 [PMID: 23474284]*

# Why do the organs fail?



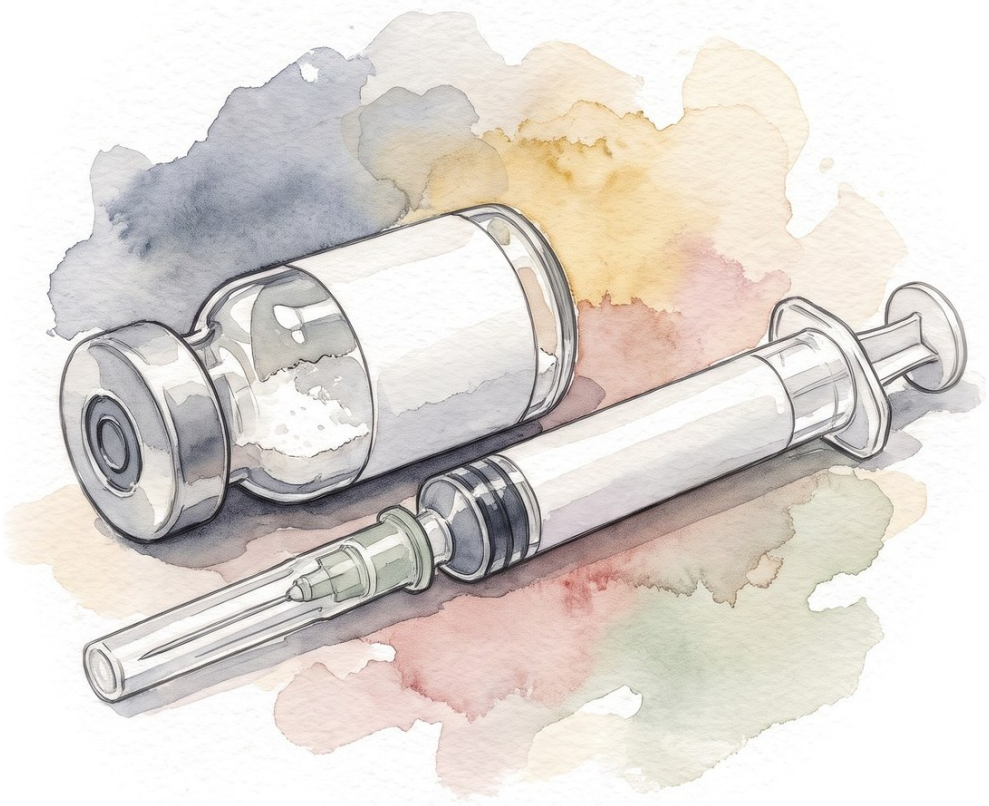

- Bed 7 so far: **kidney** and **circulation**
- Treat the trigger early and support the organs; there is no drug for the inflammation itself

*Concept from Arroyo V, Moreau R, Jalan R. N Engl J Med 2020*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

**HRS? Start terlipressin?**



# HRS-AKI: what changed in 2024

| What many of us learned (ICA 2015) | Now (ADQI-ICA 2024) |
|---|---|
| No response to 2 days off diuretics + albumin<br>No shock<br>No nephrotoxins, no structural kidney disease | Cirrhosis with ascites<br>AKI by creatinine or urine output<br>No response to 24 h of volume, if needed<br>No albumin challenge if euvolemic or wet<br>No other main cause |

**Faster to call. But you still have to look for the other causes**

*Angeli P et al. J Hepatol 2015 (ICA); ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

| **HRS-AKI (ADQI-ICA 2024)** | **Bed 7, day 1** | |
|---|---|---|
| Cirrhosis with ascites | | |
| AKI by creatinine or urine output | | |
| No response to 24 h of volume, if needed | | |
| No other main cause (shock, toxins, structural) | | |

**Go down the list for him. Which boxes does he tick?**

*ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Does he have HRS?

*12 hours later: creatinine 3.0, urine 10 mL/h, still on norepinephrine*

| **HRS-AKI (ADQI-ICA 2024)** | **Bed 7, day 1** | |
|---|---|---|
| Cirrhosis with ascites | Yes | ✅ |
| AKI by creatinine or urine output | Cr 0.9 → 3.0, urine 10 mL/h | ✅ |
| No response to 24 h of volume, if needed | 2 L on arrival; too early | ❌ |
| No other main cause (shock, toxins, structural) | **Septic shock on norepinephrine** | ❌ |

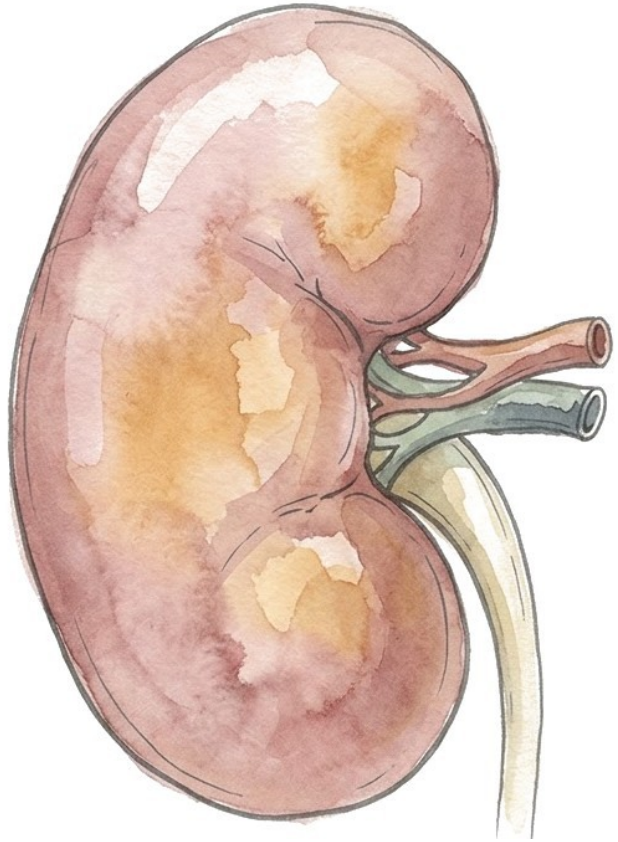
**Not HRS yet. Treat the shock first**

*ADQI-ICA consensus, J Hepatol 2024 [PMID: 38527522]*

# Kidney failure in cirrhosis is usually a mixture

**Possible contributors**

- Low effective volume
- Tubular injury: sepsis, contrast, drugs
- HRS physiology
- Tense ascites (abdominal compartment)
- Congestion

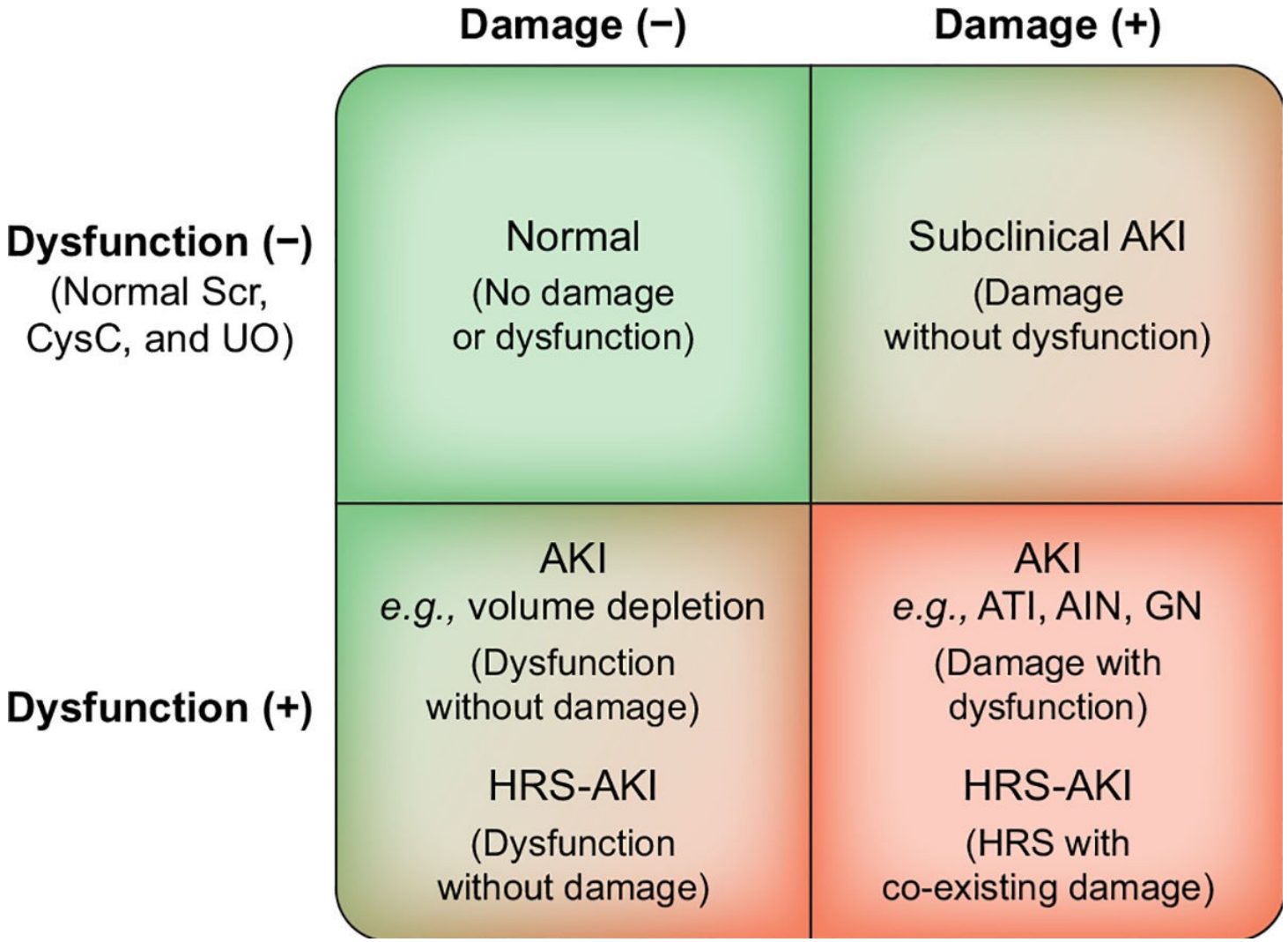


**Look**

- Urinalysis and sediment
- Drug list
- Ultrasound: kidneys, bladder, IVC, heart, lungs
- Bladder pressure if the belly is tight
- A low urine sodium is **not** proof of HRS

*ADQI-ICA consensus 2024 [PMID: 38527522]*

# Function and damage: four kinds of AKI




*Nadim MK et al. J Hepatol 2024;81:163-83 [PMID: 38527522], Fig. 4 (CC BY 2.0, adapted from ADQI)*

# Day 1: more confused

- More disoriented, pulling at his lines, asterixis. Awake, follows commands
- The nurse sent an ammonia: **88 µmol/L**

**More lactulose? Repeat the ammonia in 6 hours?**



# Day 1: more confused

- More disoriented, pulling at his lines, asterixis. Awake, follows commands
- The nurse sent an ammonia: **88 µmol/L**

**Look for the reason first. The ammonia doesn't tell you**

- Glucose, sodium, oxygen and CO2
- Is the infection controlled? Cultures, source, antibiotic coverage
- Bleeding, sedatives and opioids, withdrawal, seizures, a new focal deficit (CT head)
- Lactulose for overt HE: 2 to 3 soft stools a day. By tube if he can't swallow safely; watch for dehydration and high sodium
- Rifaximin is an add-on, mainly to prevent the next episode
- Grade III-IV and a poor airway: intubate
- No repeat ammonia

*EASL HE guideline 2022 [PMID: 35724930]; SCCM 2023 [PMID: 37052436]*

# Day 2: worse kidney, worse lungs

| | Admission | Day 2 |
|---|---|---|
| Albumin received | | 110 g (SBP dose, 1.5 g/kg at 73 kg) + 2 L crystalloid |
| Breathing | SpO2 92%, Venturi 35% | **SpO2 86% on Venturi 50%, RR 32, crackles** |
| Ultrasound | | Diffuse B-lines, IVC full |
| Creatinine (mg/dL) | 2.6 | 3.1 |
| Urine output | 15 mL/h | 10 mL/h |
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 |

**More albumin? Terlipressin? Something else?**

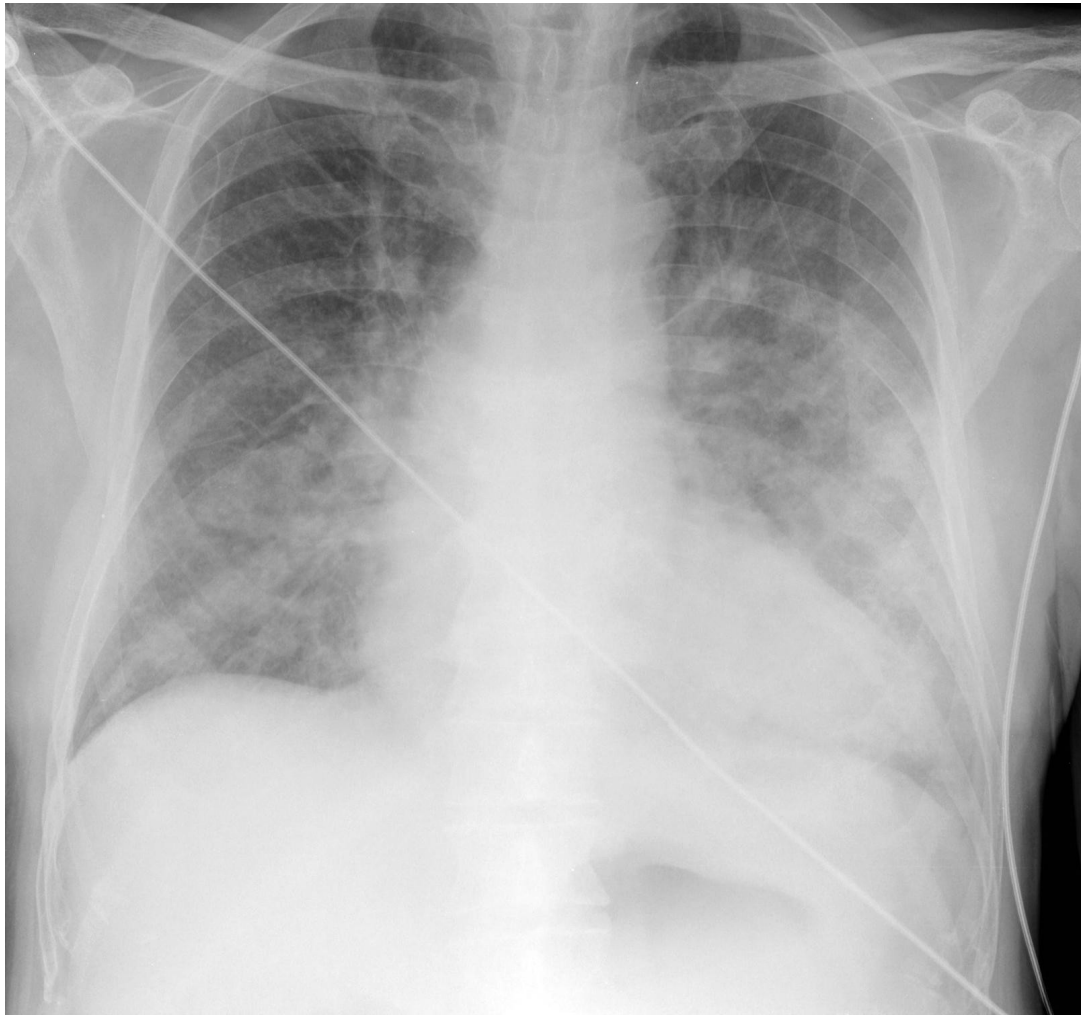
# Day 2: worse kidney, worse lungs

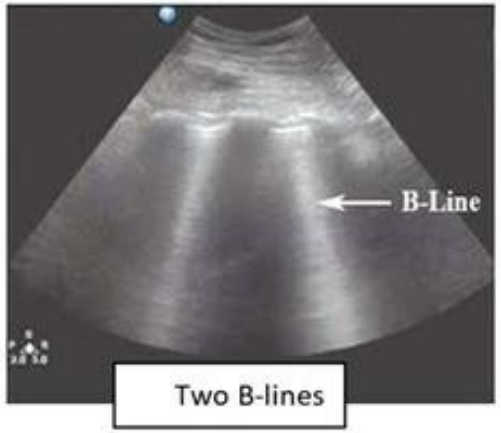
| | Admission | Day 2 |
|---|---|---|
| Albumin received | | 110 g (SBP dose, 1.5 g/kg at 73 kg) + 2 L crystalloid |
| Breathing | SpO2 92%, Venturi 35% | **SpO2 86% on Venturi 50%, RR 32, crackles** |
| Ultrasound | | Diffuse B-lines, IVC full |
| Creatinine (mg/dL) | 2.6 | 3.1 |
| Urine output | 15 mL/h | 10 mL/h |
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 |

## First, what are the lungs telling you?

**Now on high-flow, FiO2 0.6, SpO2 90%: SpO2/FiO2 150. That's a third organ failure**

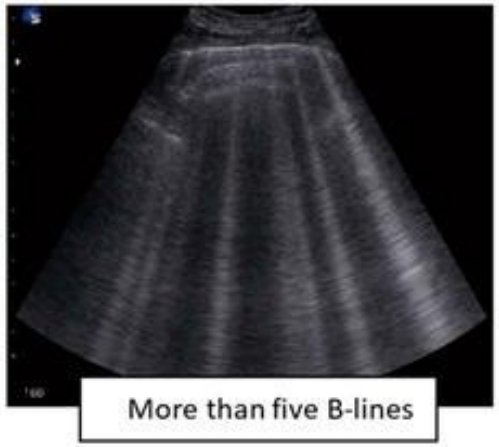
# Day 2: what do his lungs look like?

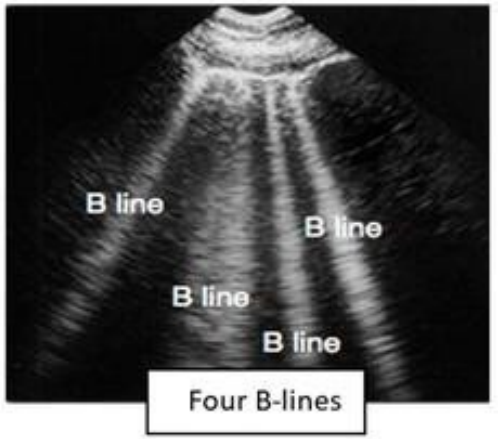








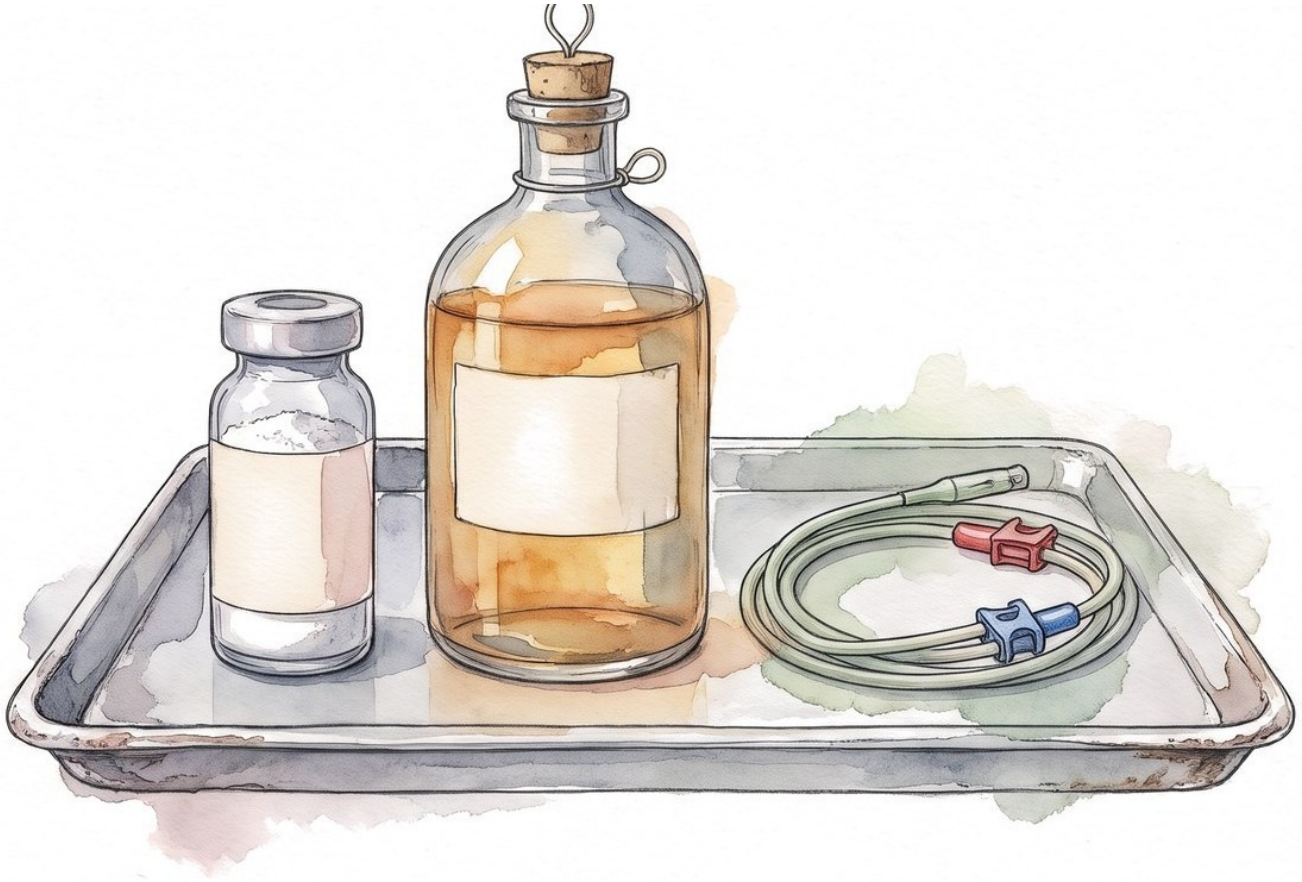



*Images for teaching, not this patient*

What would you do for his kidney today?

**Terlipressin, more albumin, or dialysis?**



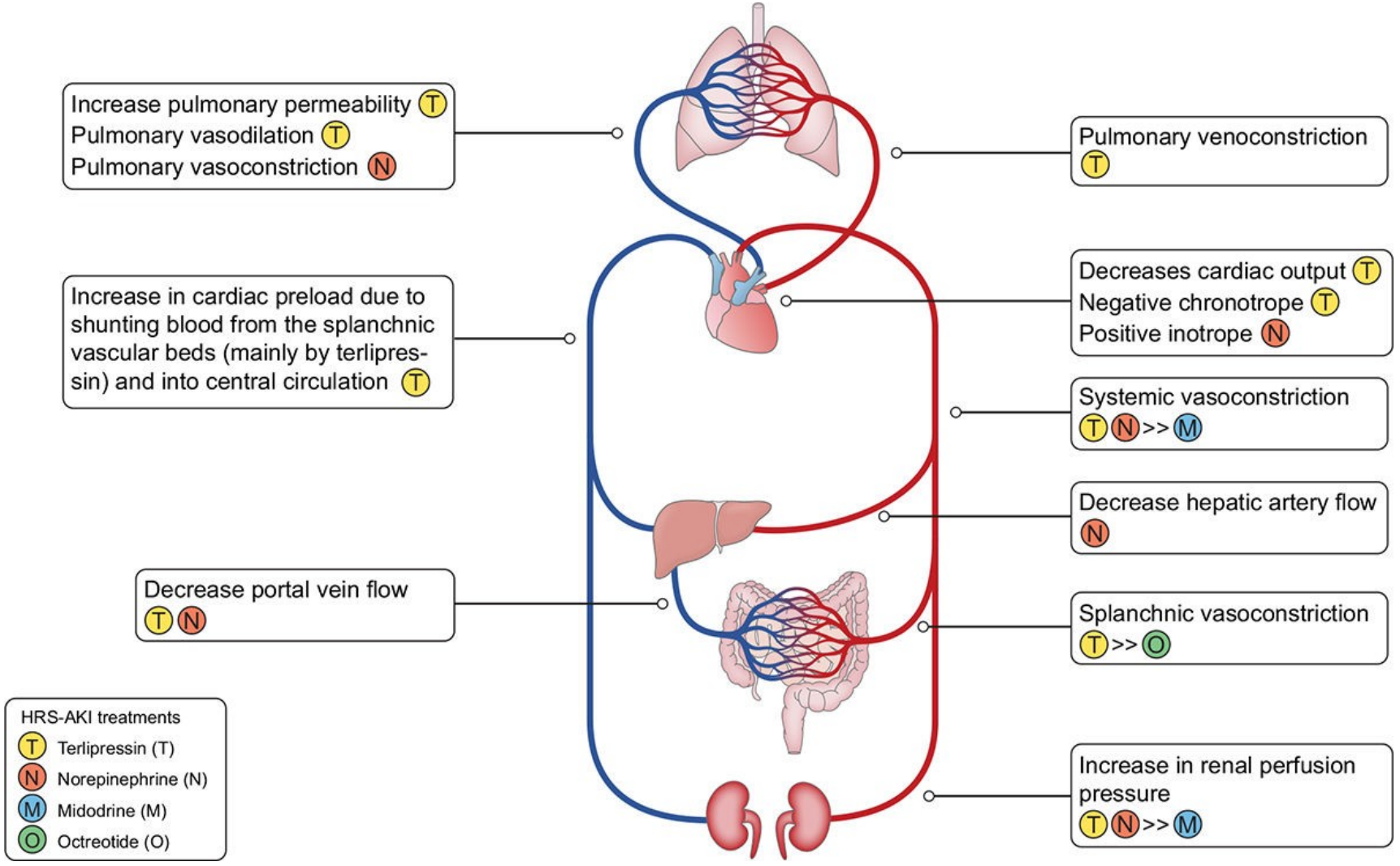
# What would you do for his kidney today?

- **Terlipressin: not for him today**
  - Label: don't start if SpO2 < 90% or breathing is getting worse
  - Label: avoid in ACLF grade 3 and in volume overload
- Stop the albumin while he is wet
- He is already on a vasoconstrictor. Norepinephrine, titrated to perfusion
- RRT if he stays overloaded. For HRS, RRT is a bridge to transplant: if he isn't a candidate, talk about goals before you start

**Check oxygen and volume before any kidney treatment**

*TERLIVAZ prescribing information, revised 6/2026; ADQI-ICA 2024 [PMID: 38527522]*

# What each HRS drug does to the heart and lungs




*Nadim MK et al. J Hepatol 2024;81:163-83 [PMID: 38527522], Fig. 5 (CC BY 2.0, adapted from ADQI)*

# Does HRS reversal mean survival?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

**What do you think they found?**

*Wong F et al. N Engl J Med 2021;384:818-28 [PMID: 33657294]*

# Does HRS reversal mean survival?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## Terlipressin plus Albumin for the Treatment of Type 1 Hepatorenal Syndrome

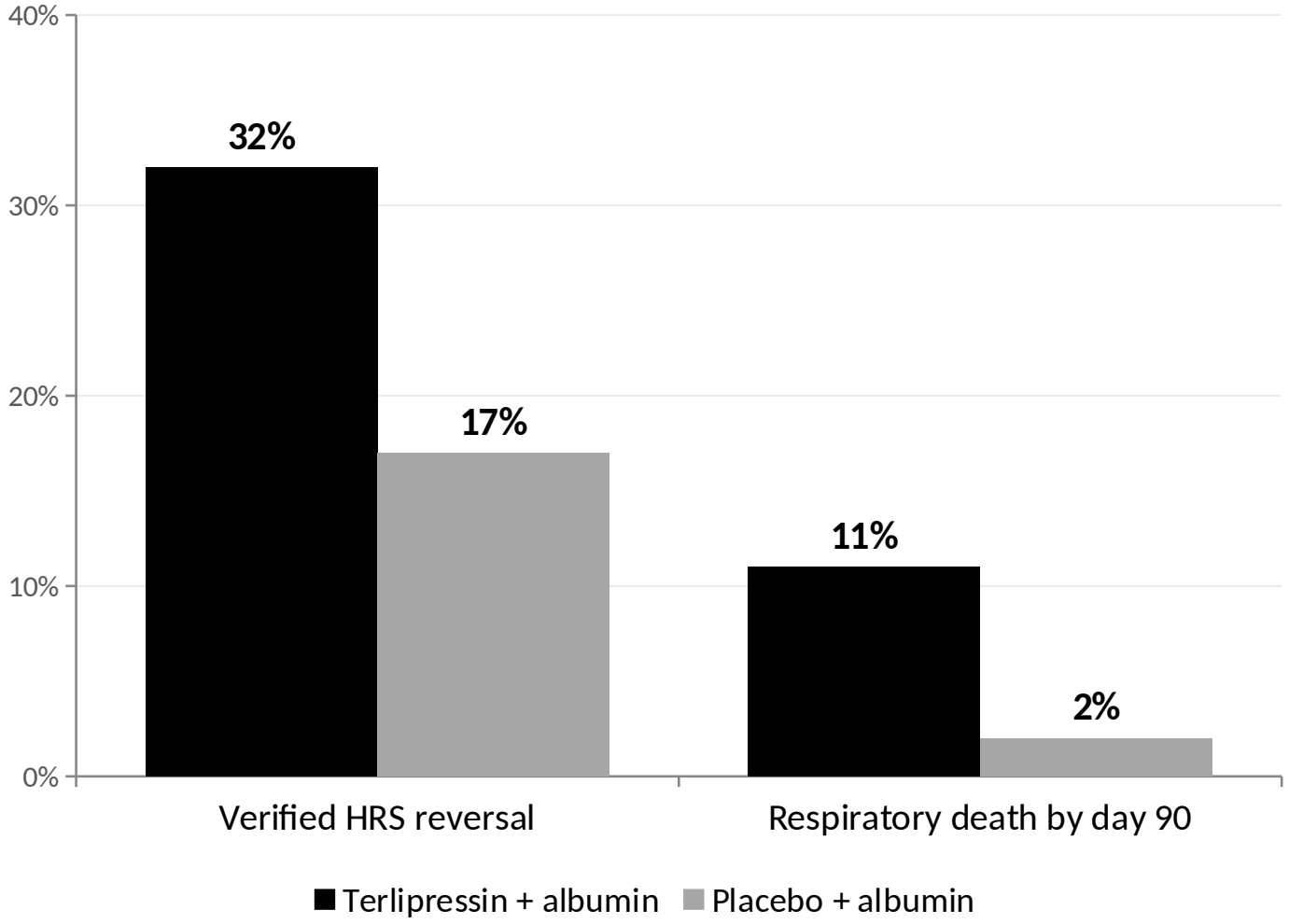
F. Wong, S.C. Pappas, M.P. Curry, K.R. Reddy, R.A. Rubin, M.K. Porayko, S.A. Gonzalez, K. Mumtaz, N. Lim, D.A. Simonetto, P. Sharma, A.J. Sanyal, M.J. Mayo, R.T. Frederick, S. Escalante, and K. Jamil, for the CONFIRM Study Investigators*

**CONCLUSIONS**

In this trial involving adults with cirrhosis and HRS-1, terlipressin was more effective than placebo in improving renal function but was associated with serious adverse events, including respiratory failure. (Funded by Mallinckrodt Pharmaceuticals; CONFIRM ClinicalTrials.gov number, NCT02770716.)

*Wong F et al. N Engl J Med 2021;384:818-28 [PMID: 33657294]*

# Does HRS reversal mean survival?




**CONFIRM: type 1 HRS, 300 patients**

- More kidneys recovered
- **More patients died of respiratory failure**
- All-cause 90-day mortality was not different
- Respiratory failure as an adverse event: about 14% vs 5% (label)

*Wong F et al. NEJM 2021 [PMID: 33657294]; TERLIVAZ prescribing information, revised 6/2026*

# Should we treat the albumin number?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## A Randomized Trial of Albumin Infusions in Hospitalized Patients with Cirrhosis

Louise China, Ph.D., Nick Freemantle, Ph.D., Ewan Forrest, M.D., Yiannis Kallis, Ph.D., Stephen D. Ryder, D.M., Gavin Wright, Ph.D., Andrew J. Portal, M.D., Natalia Becares Salles, Ph.D., Derek W. Gilroy, Ph.D., and Alastair O'Brien, Ph.D., for the ATTIRE Trial Investigators*

**What do you think they found?**

*China L et al. N Engl J Med 2021;384:808-17 [PMID: 33657293]*

# Should we treat the albumin number?

The NEW ENGLAND JOURNAL of MEDICINE

ORIGINAL ARTICLE

## A Randomized Trial of Albumin Infusions in Hospitalized Patients with Cirrhosis

Louise China, Ph.D., Nick Freemantle, Ph.D., Ewan Forrest, M.D., Yiannis Kallis, Ph.D., Stephen D. Ryder, D.M., Gavin Wright, Ph.D., Andrew J. Portal, M.D., Natalia Becares Salles, Ph.D., Derek W. Gilroy, Ph.D., and Alastair O'Brien, Ph.D., for the ATTIRE Trial Investigators*

**CONCLUSIONS**

In patients hospitalized with decompensated cirrhosis, albumin infusions to increase the albumin level to a target of 30 g per liter or more was not more beneficial than the current standard care in the United Kingdom.

*China L et al. N Engl J Med 2021;384:808-17 [PMID: 33657293]*

# Should we treat the albumin number?

| | |
|---|---|
| **Study** | 777 inpatients: albumin to ≥ 30 g/L vs standard care |
| **Finding** | Infection, kidney dysfunction or death: 29.7% vs 30.2% |
| **The catch** | **More serious adverse events, including pulmonary edema** |

**When albumin is indicated**

- SBP: 1.5 g/kg on day 1, 1 g/kg on day 3
- Paracentesis over 5 L: 6 to 8 g per liter removed
- HRS-AKI: with a vasoconstrictor, dosed to volume status

*China L et al. NEJM 2021 [PMID: 33657293]; ADQI-ICA 2024 [PMID: 38527522]*

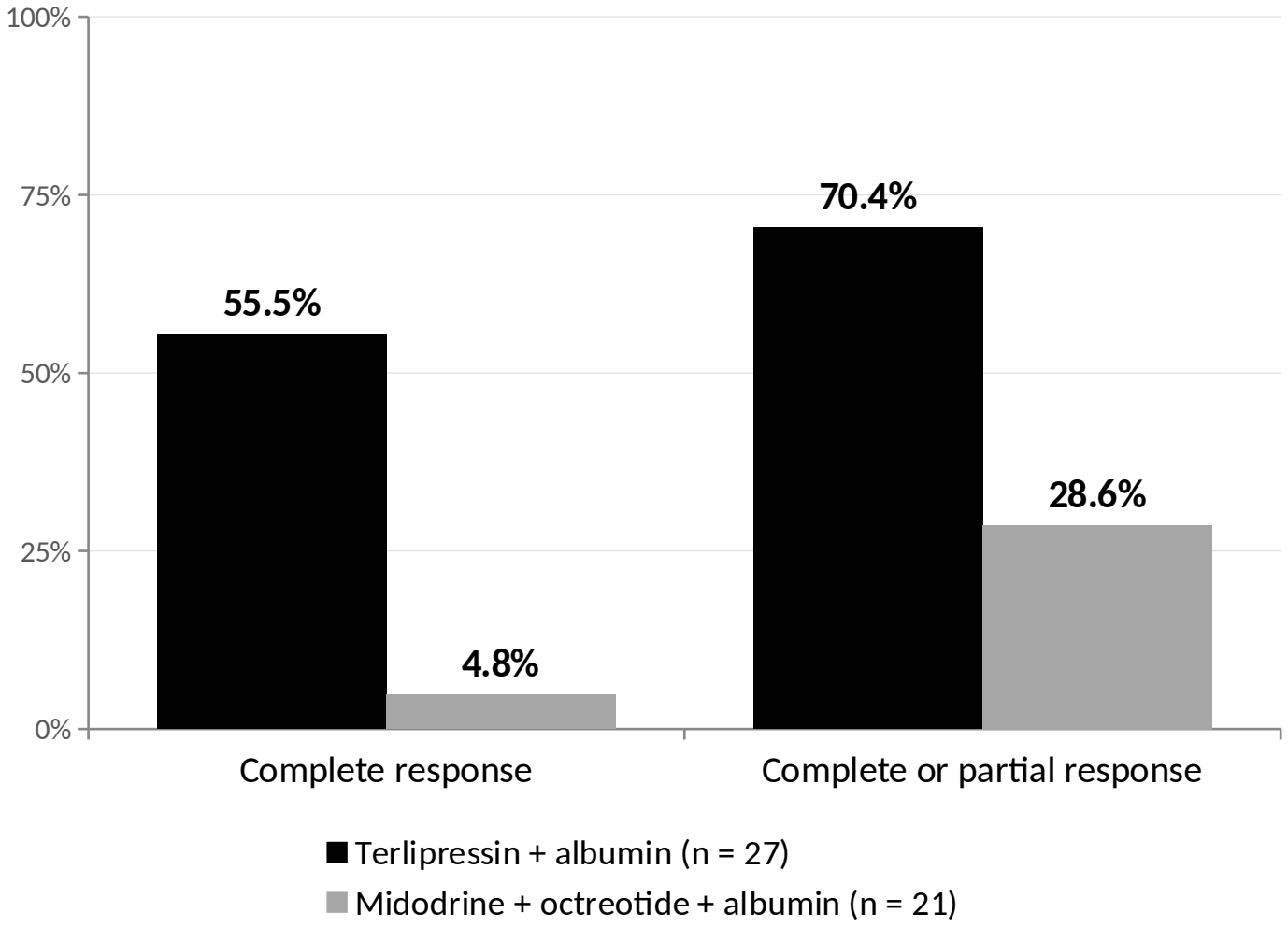
# Midodrine and octreotide for HRS-AKI?

Head to head: terlipressin vs midodrine plus octreotide, both with albumin, type 1 HRS

## How different can two vasoconstrictors be?



# Midodrine and octreotide for HRS-AKI?




**Randomized, 8 Italian centers, type 1 HRS. 49 randomized (27 vs 22), 48 analyzed**

- Stopped early at the interim analysis because the difference was already clear
- Blood pressure rose more with terlipressin
- 3-month survival 59% vs 43%; too small to tell
- Not a substitute for IV pressors in the ICU
- **Use it only if terlipressin and norepinephrine are not options**

*Cavallin M et al. Hepatology 2015;62:567-74 [PMID: 25644760]; ADQI-ICA 2024 [PMID: 38527522]*

# What changed since yesterday?

| | Day 1 | Day 2 | Day 3 | Day 5 |
|---|---|---|---|---|
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 | 0.12 | Off |
| Oxygen, SpO2 | Venturi 0.35, 92% | HFNC 0.6, 90% | HFNC 0.4, 95% | 2 L, 97% |
| Lung failure? | No (P/F 211) | Yes (S/F 150) | No (S/F 238) | No |
| Kidney | Cr 2.6 | Cr 3.1 | CRRT for fluid | On CRRT |
| Encephalopathy | Grade II | Grade II | Grade I | Grade I |
| Bilirubin / INR | 5.8 / 2.1 | 6.4 / 2.3 | 6.1 / 2.2 | 5.2 / 2.0 |
| ACLF grade | 2 | **3** | 2 | ? |

**Better or worse? What do you tell his family?**

*Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]*

# What changed since yesterday?

| | Day 1 | Day 2 | Day 3 | Day 5 |
|---|---|---|---|---|
| Norepinephrine (mcg/kg/min) | 0.15 | 0.22 | 0.12 | Off |
| Oxygen, SpO2 | Venturi 0.35, 92% | HFNC 0.6, 90% | HFNC 0.4, 95% | 2 L, 97% |
| Lung failure? | No (P/F 211) | Yes (S/F 150) | No (S/F 238) | No |
| Kidney | Cr 2.6 | Cr 3.1 | CRRT for fluid | On CRRT |
| Encephalopathy | Grade II | Grade II | Grade I | Grade I |
| Bilirubin / INR | 5.8 / 2.1 | 6.4 / 2.3 | 6.1 / 2.2 | 5.2 / 2.0 |
| ACLF grade | 2 | **3** | 2 | **1** |

- The grade at days 3 to 7 predicts outcome better than the grade at admission
- Persistent ACLF 3 at days 3 to 7, especially with four or more organ failures, has a very high mortality without transplant
- If he is not a transplant candidate, discuss goals of care with the family and involve palliative care

*Made-up numbers for teaching. CLIF-C ACLF score [PMID: 24950482]; EASL ACLF 2023 [PMID: 37364789]*

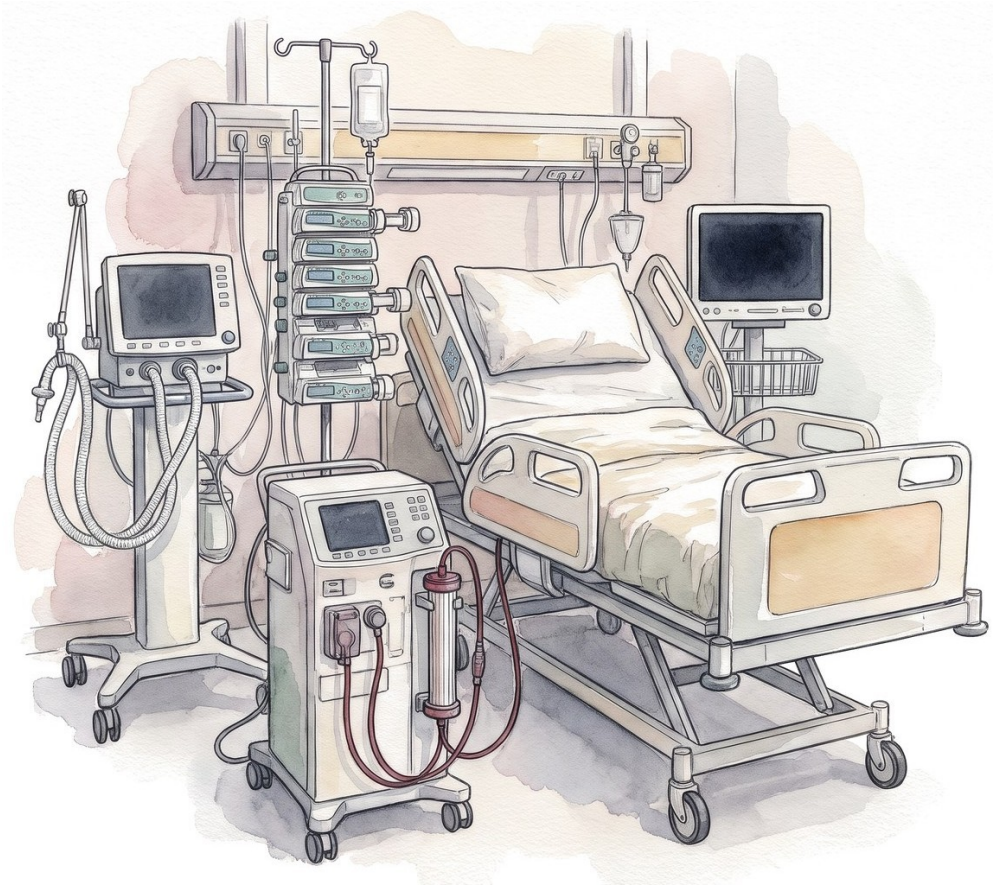
# Who gets a liver?

Transplant in ACLF



# Too sick for a transplant, or sick enough to need one?

**Ventilator, pressors, CRRT, ACLF 3. Is he too sick?**



# Too sick for a transplant, or sick enough to need one?

- A ventilator, pressors or dialysis do not by themselves exclude transplant
- Selected patients with ACLF grade 3 do well after transplant. The window is short

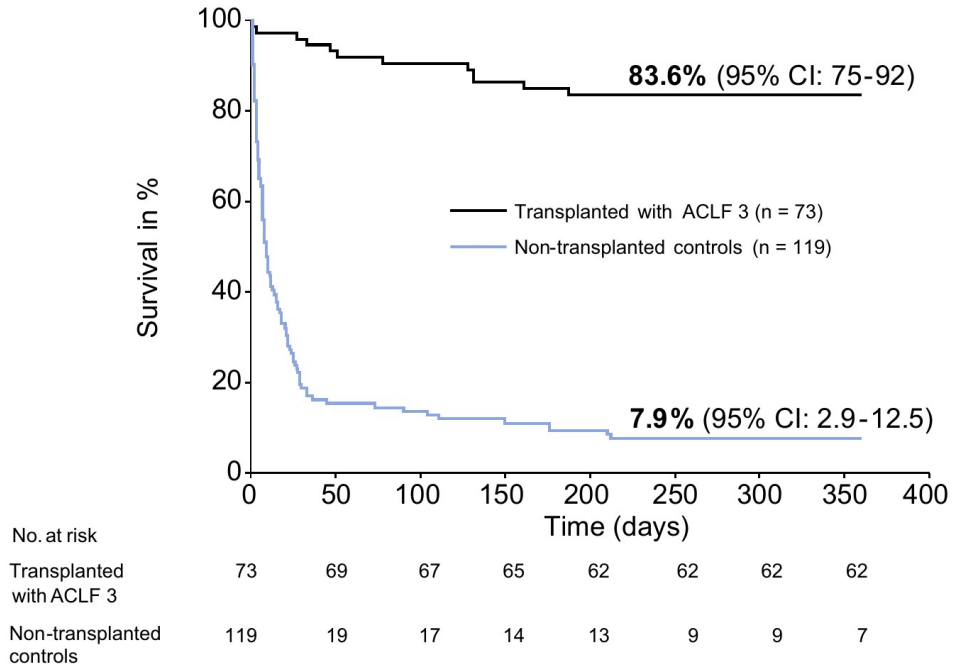
**What can close the window**

- Uncontrolled infection
- Rising vasopressor needs, severe respiratory failure
- Irreversible brain injury
- Frailty and comorbidity
- Alcohol: centers now evaluate individually rather than by a fixed abstinence rule

Research Article

EASL JOURNAL OF HEPATOLOGY

**Liver transplantation in the most severely ill cirrhotic patients: A multicenter study in acute-on-chronic liver failure grade 3**




*Selected patients, not randomized*

*Artru F et al. J Hepatol 2017 [PMID: 28645736]; Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]; AASLD/AST 2026 [PMID: 41405234]*

# Calling transplant hepatology

**Call first. Gather this while you wait for the call back**

| Patient A (ALF) | Patient B (ACLF) |
|---|---|
| Cause, days from jaundice to HE | Cause of cirrhosis, recent alcohol use |
| Encephalopathy grade, pupils, sedation | Precipitant and source control |
| Airway and ventilator settings | Organ failures today and yesterday |
| Ammonia, sodium, CRRT | Pressor dose, oxygen, RRT |
| INR, lactate, pH, glucose trends | MELD, CLIF-C scores |
| Pressors, infection | Frailty, comorbidities |
| Social support, intent of ingestion | Prior transplant evaluation, support |

# ALF and ACLF: what changes at the bedside

*Both have a sick liver. Do they have the same disease?*

| | Acute liver failure | ACLF |
|---|---|---|
| **Ammonia** | | |
| **Lactulose** | | |
| **Sodium** | | |
| **Antibiotics** | | |
| **INR** | | |

*ACG ALF 2023 [PMID: 37377263]; SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]*

# ALF and ACLF: what changes at the bedside

*Both have a sick liver. Do they have the same disease?*

| | Acute liver failure | ACLF |
|---|---|---|
| **Ammonia** | With HE grade and trend: brain risk, CRRT | Doesn't diagnose HE; don't trend it |
| **Lactulose** | No proven benefit; distends the bowel | Yes for overt HE ± rifaximin |
| **Sodium** | HE III-IV: often 145-155, match CRRT | Often low: ≤ 8 mmol/L per 24 h |
| **Antibiotics** | No routine prophylaxis; low threshold | Treat fast; prophylaxis with GI bleed |
| **INR** | Tracks severity, not bleeding | Same; judge the procedure |

*ACG ALF 2023 [PMID: 37377263]; SCCM 2023 [PMID: 37052436]; update 2022 [PMID: 36001996]*

# What happened to them

## Bed 4



**Day 0**
Admitted, NAC

**Day 1**
Intubated, CRRT

**Day 2**
Listed 1A

**Day 4**
Transplanted

**Day 19**
Home

## Bed 7



**Day 1**
SBP, ACLF 2

**Day 2**
ACLF 3

**Day 5**
Off pressors, ACLF 1

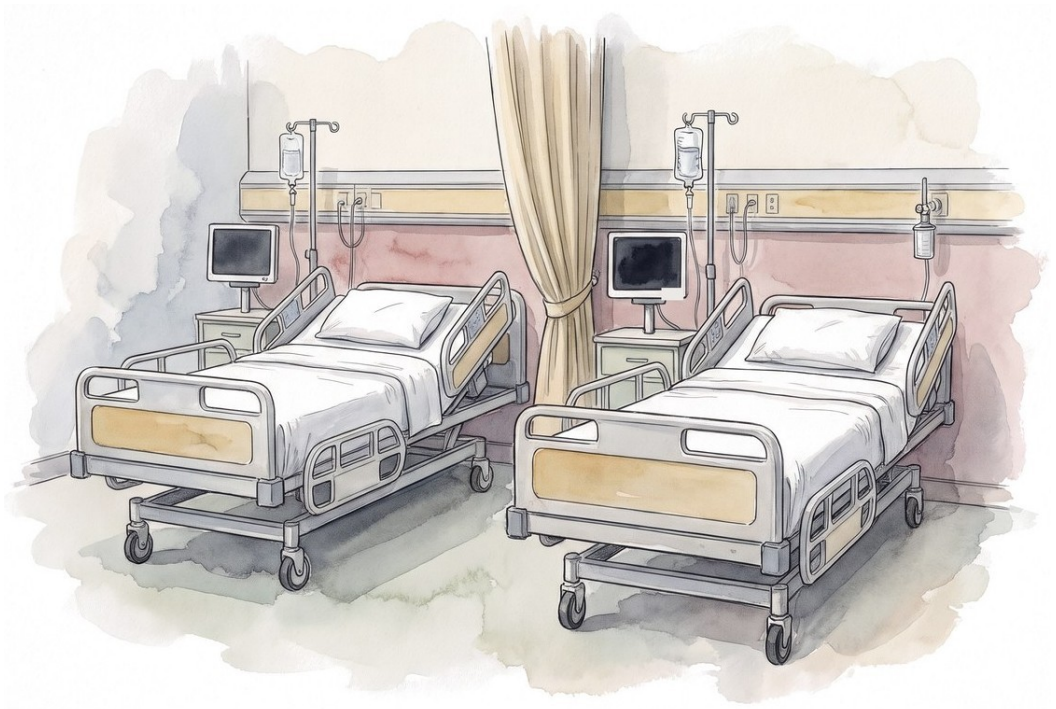
**Day 12**
Off CRRT

**Day 21**
Home, transplant clinic

*Made-up outcomes for teaching*

# Back to our patients

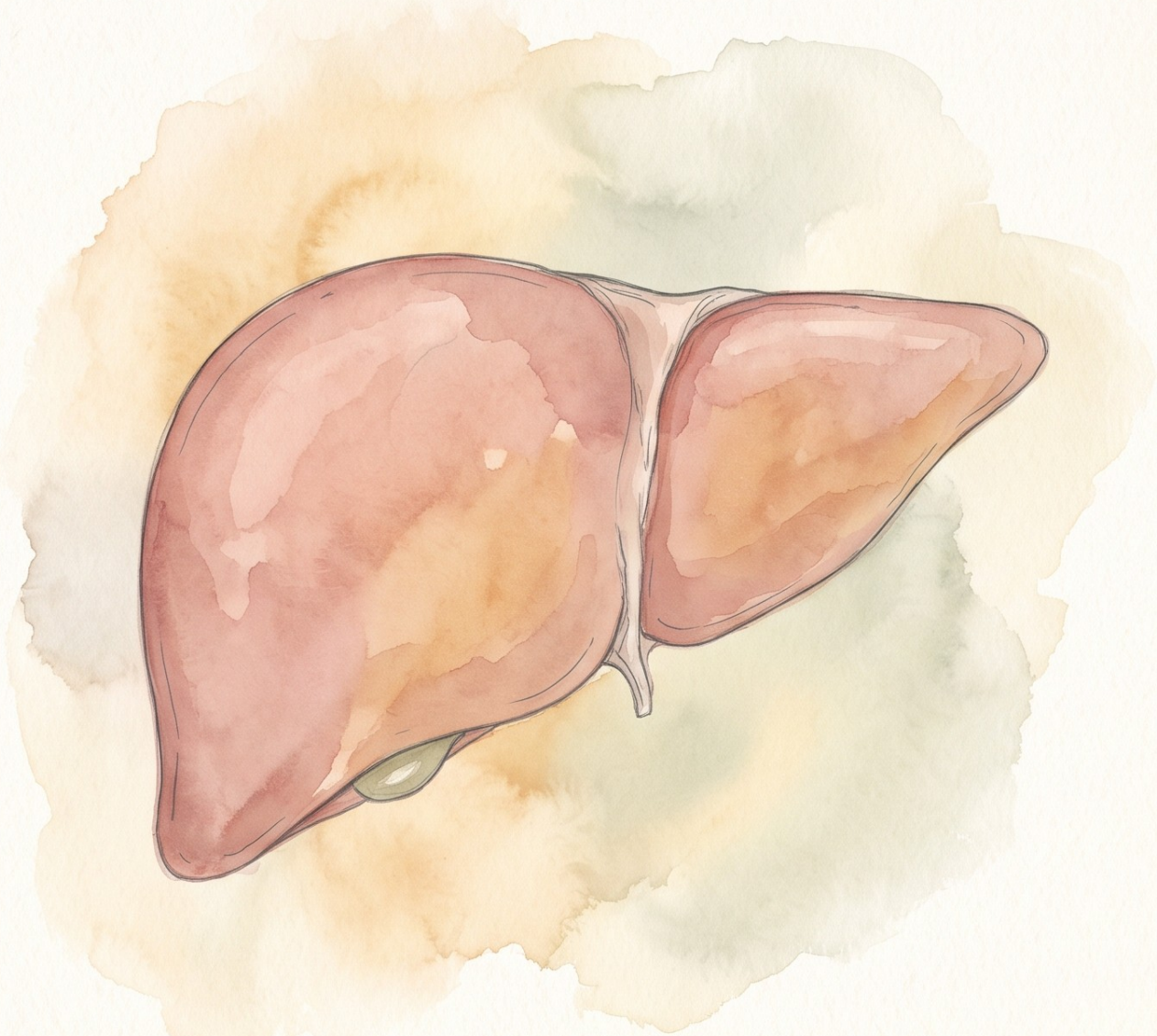
- Which one needed transplant hepatology before the tests came back?
- Why did a falling ALT make you more worried?
- When is CRRT right before the creatinine is high?
- Why isn't lactulose for ACLF evidence for ALF?
- Why were albumin and terlipressin the wrong next step for bed 7?
- What would make you change yesterday's plan?



# Some tricks of the trade

- High AST and ALT alone are not liver failure. Get an INR and examine the patient
- Stabilize, treat the cause and call transplant hepatology in the same hour
- ALF: **protect the brain**. Intubate at grade III, keep the CO2 normal, follow your center's sodium plan, and start CRRT for rising ammonia
- ACLF: count the organs, find the trigger
- ACLF: check oxygenation and volume status before albumin or terlipressin
- Reassess organ failures daily. The trend matters more than the admission score





# Questions?

# Appendix: West Haven grades

| Grade | What you see |
|---|---|
| I | Short attention, mild confusion, reversed sleep, subtle personality change |
| II | Lethargy, disorientation to time, obvious personality change, asterixis |
| III | Somnolent but rousable, grossly disoriented, bizarre behavior |
| IV | Coma |

# Appendix: King's College criteria

| Acetaminophen | Not acetaminophen |
|---|---|
| **Arterial pH < 7.30 after fluids**<br>*or all three within 24 hours:*<br>INR > 6.5<br>Creatinine > 3.4 mg/dL<br>Grade III-IV encephalopathy<br>*(Modified: lactate > 3.5 at 4 h or > 3.0 at 12 h after fluids)* | **INR > 6.5**<br>*or any three of:*<br>Age < 10 or > 40<br>Drug reaction or unknown cause<br>Jaundice to encephalopathy > 7 days<br>INR > 3.5<br>Bilirubin > 17.5 mg/dL |

*Original O'Grady criteria; the lactate line is the later modification*

*O'Grady 1989 criteria, as summarized in EASL 2017 [PMID: 28417882] and update 2022 [PMID: 36001996]*

# Appendix: arterial ammonia studies in ALF

| Study (arterial, µmol/L) | Patients | What they found | The catch |
|---|---|---|---|
| Clemmesen 1999 | 44 ALF | **Herniated: mean 230 vs 118; all above 146** | Small, one center |
| Bhatia 2006 | 80 ALF | ≥ 124 on admission predicted death | Endpoint was death, not ICH |
| Bernal 2007 | 165, HE III-IV | > 100: severe HE. **> 200: ICH in 55%** | 200 missed most ICH |
| Kumar 2012 | 295 ALF | **≥ 122 for 3 days: survival 23% vs 72%** | Trend beat a single value |

**Above 150 to 200 and not falling: worry about the brain**

*ICH can occur below 200. Interpret the level with the HE grade and the trend*

*Clemmesen 1999 [PMID: 10051463]; Bhatia 2006 [PMID: 16024550]; Bernal 2007 [PMID: 17685471]; Kumar 2012 [PMID: 22521861]*

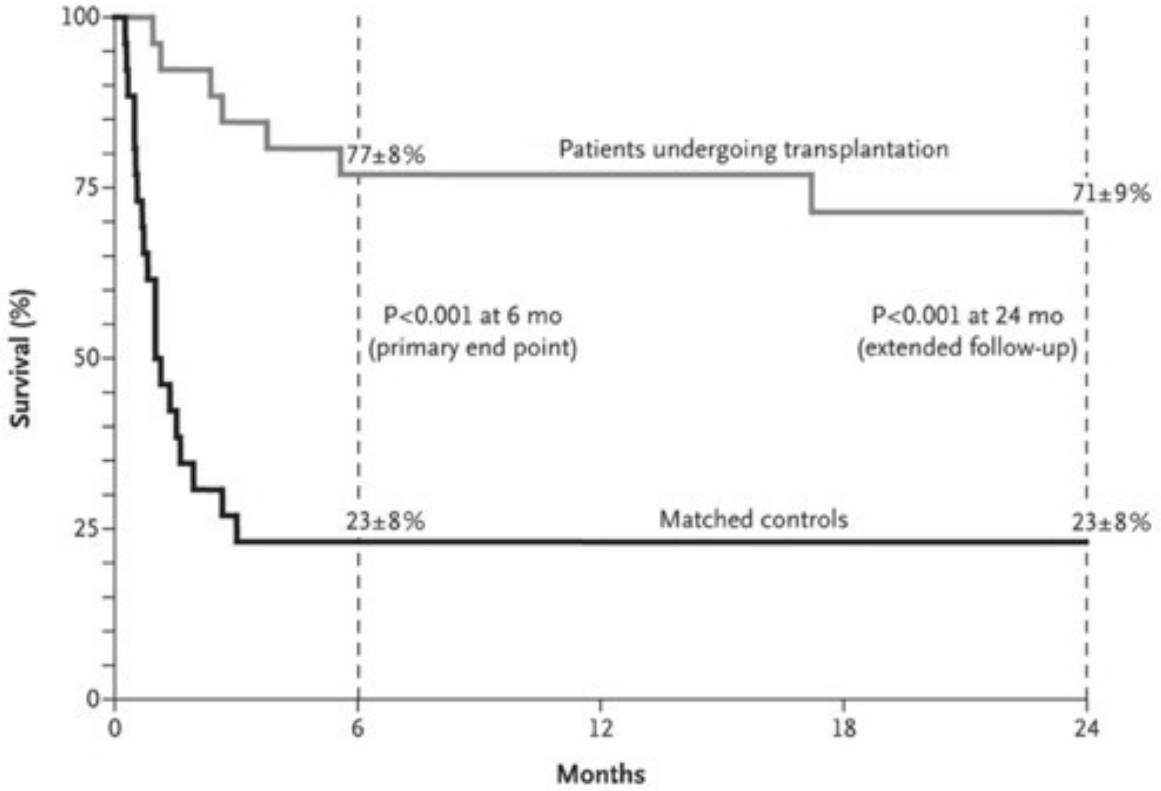
# Appendix: which ACLF?

| | EASL-CLIF | NACSELD | APASL |
|---|---|---|---|
| **Who** | Decompensated cirrhosis | Cirrhosis, inpatient | CLD ± cirrhosis |
| **Defined by** | Organ failures (CLIF-C) | ≥ 2 extrahepatic failures | Liver insult: jaundice + INR ≥ 1.5 |
| **Picks up** | Broader, earlier | Sicker patients | A primary liver insult |

**They find different patients. Say which one you're using**

*Kyoto consensus 2025 [PMID: 39961976]; ACG ACLF 2022 [PMID: 35006099]; AASLD 2024 [PMID: 37939273]*

# Appendix: early transplant for alcohol-associated hepatitis




| No. at Risk | | | | | |
|---|---|---|---|---|---|
| Patients undergoing transplantation | 26 | 20 | 15 | 14 | 13 |
| Matched controls | 26 | 6 | 6 | 5 | 4 |

- 26 patients with a first episode of severe alcohol-associated hepatitis, not responding to steroids
- Highly selected: first liver-related event, close family support, no severe other illness, agreed to abstinence
- 6-month survival **77% vs 23%**
- Recent drinking alone is not a reason to skip the call

*Mathurin P et al. N Engl J Med 2011 [PMID: 22070476]*

# Appendix: sources (1)


- SCCM ALF/ACLF ICU guidelines, part 1. Crit Care Med 2020 [PMID: 32058387]
- SCCM ALF/ACLF ICU guidelines, part 2. Crit Care Med 2023 [PMID: 37052436]
- EASL acute liver failure guideline. J Hepatol 2017 [PMID: 28417882]
- ACG acute liver failure guideline. Am J Gastroenterol 2023 [PMID: 37377263]
- Management of acute liver failure: update 2022. Semin Liver Dis [PMID: 36001996]
- Acute liver failure: a practical update. JHEP Rep 2024 [PMID: 39170946]
- EASL ACLF guideline. J Hepatol 2023 [PMID: 37364789]
- AASLD ACLF and critically ill cirrhosis guidance. Hepatology 2024 [PMID: 37939273], errata [PMID: 38551831], [PMID: 42467893]
- ACG ACLF guideline. Am J Gastroenterol 2022 [PMID: 35006099]
- ADQI-ICA consensus on AKI in cirrhosis. J Hepatol 2024 [PMID: 38527522]
- Surviving Sepsis Campaign 2026. Crit Care Med [PMID: 41869847]
- EASL autoimmune hepatitis guideline. J Hepatol 2025 [PMID: 40348684]
- EASL hepatic encephalopathy guideline. J Hepatol 2022 [PMID: 35724930]
- US/Canada acetaminophen poisoning consensus. JAMA Netw Open 2023 [PMID: 37552484]

# Appendix: sources (2)


- Moreau R et al. CANONIC. Gastroenterology 2013 [PMID: 23474284]
- Jalan R et al. CLIF-C ACLF score. J Hepatol 2014 [PMID: 24950482]
- Lee WM et al. NAC in non-acetaminophen ALF. Gastroenterology 2009 [PMID: 19524577]
- Clemmesen JO et al. Cerebral herniation and arterial ammonia. Hepatology 1999 [PMID: 10051463]
- Bernal W et al. Arterial ammonia, HE and intracranial hypertension. Hepatology 2007 [PMID: 17685471]
- Bhatia V et al. Arterial ammonia in ALF. Gut 2006 [PMID: 16024550]
- Kumar R et al. Persistent hyperammonemia in ALF. Clin Gastroenterol Hepatol 2012 [PMID: 22521861]
- Cardoso FS et al. CRRT and ammonia in ALF. Hepatology 2018 [PMID: 28859230]
- Pre-emptive CRRT in ALF with cerebral edema. Aliment Pharmacol Ther 2026 [PMID: 42332991]
- Larsen FS et al. High-volume plasma exchange. J Hepatol 2016 [PMID: 26325537]
- Cavallin M et al. Terlipressin vs midodrine and octreotide. Hepatology 2015 [PMID: 25644760]
- Wong F et al. CONFIRM. NEJM 2021 [PMID: 33657294]
- Artru F et al. Transplant in ACLF grade 3. J Hepatol 2017 [PMID: 28645736]
- Mathurin P et al. Early transplant for alcoholic hepatitis. NEJM 2011 [PMID: 22070476]
- China L et al. ATTIRE. NEJM 2021 [PMID: 33657293]
- Liver transplantation for ACLF. Lancet Gastroenterol Hepatol 2024 [PMID: 38309288]
- AASLD/AST candidate evaluation guideline. Hepatology 2026 [PMID: 41405234]
- Kyoto consensus on ACLF. Hepatol Int 2025 [PMID: 39961976]
- TERLIVAZ (terlipressin) prescribing information, revised 6/2026